ARPES viewer — User Guide

中文版 / Chinese version · *Version française*

This guide walks through the program's main features with screenshots, in the order you would normally use them: loading data, looking at it, converting it to momentum, analysing it, and making figures. The technical background for each step (formulas, test results, design choices) is in the README and CHANGELOG.

Every screenshot was taken from the running program, using the synthetic data written by `devtools/demo_data.py`, so you can follow along without a beamtime (see Trying it with demo data).

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1. Installing and starting

You need Python 3.9 or newer and five packages:

```
pip install PyQt5 pyqtgraph numpy h5py scipy
```

Then, from the `ARPES_viewer` folder:

```
python ARPES_viewer.py
```

The program reads:

Beamline / source	Files
SOLEIL **ANTARES**	NeXus/HDF5 `*.nxs` (cuts, deflector maps, spatial scans)
SOLEIL **CASSIOPEE**	Scienta SES text files — single spectra, and numbered folders assembled into a map or a photon-energy scan
CASSIOPEE **spin end station** (MBS A-1)	`*.krx` and `*.txt` — cuts, deflector maps and spin-resolved EDCs
This program	its own saved `*.nxs` format

2. Trying it with demo data

```
python devtools/demo_data.py            # writes ./demo_data/*.nxs
```

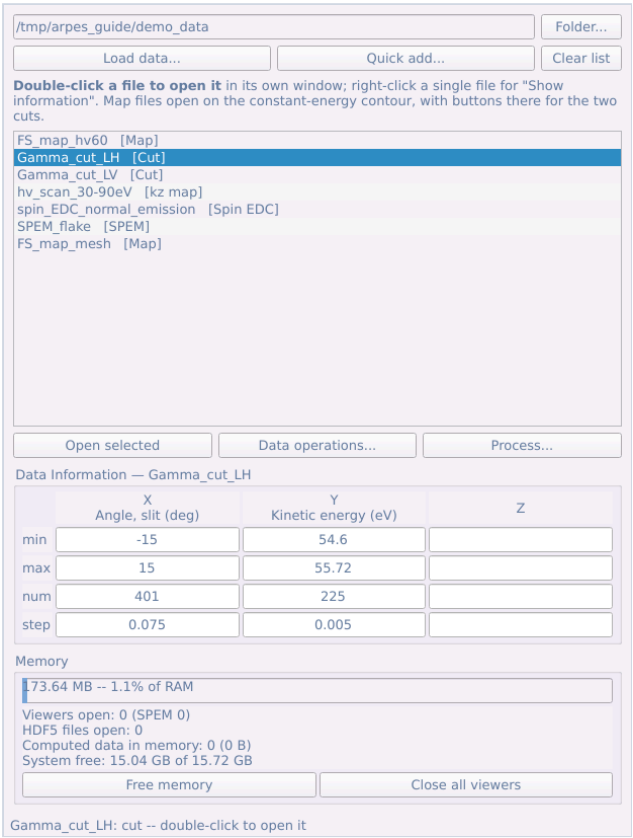
This writes six files of synthetic data: a Fermi-surface map, a pair of cuts taken with LH and LV polarisation, a photon-energy scan, a spin-resolved EDC, a small real-space scan and a map with a detector mesh printed on it. They come from a toy band structure (a hexagonal electron pocket and a hole band at Γ, hole pockets at K) with a Fermi cut-off, the analyser resolution and Poisson noise, and open exactly like measured data.

To re-take every screenshot in this guide after changing a window:

```
python devtools/guide_screenshots.py    # writes docs/images/*.png
```

3. The main window

The main window is a **browser**: one row per *dataset*, not per file (one `.nxs` file often holds several measurements, and each gets its own row). The kind is written after the name: `[Cut]`, `[Map]`, `[kz map]`, `[k-map]`, `[SPEM]`, `[EDC]`, `[Spin EDC]` …

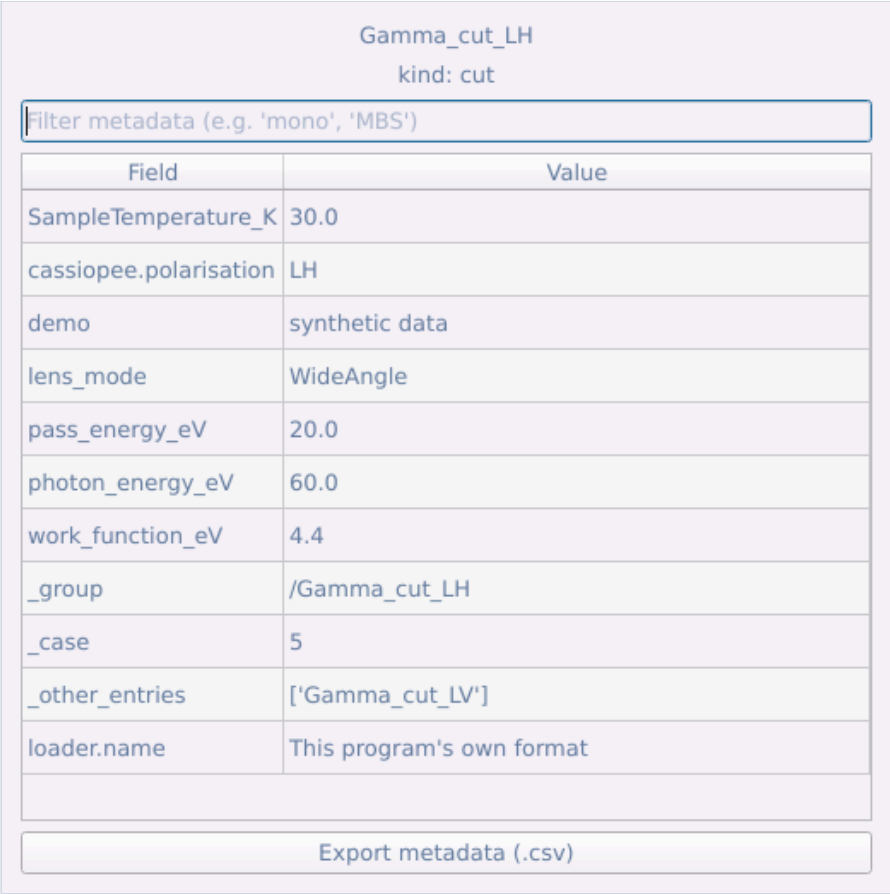


Action	How
Open a dataset in its own window	**double-click** the row, or **Open selected**
Everything else that works on datasets	**right-click** the row (or a selection of rows)
Read the axes: first/last value, points, step	right-click → **Show information** (fills *Data Information*)
Truncate / normalise / compress several datasets at once	**Data operations...**

Action	How
Smooth, differentiate, curvature, backgrounds, symmetrise	**Process...**
See and free memory	the **Memory** box at the bottom

The right-click menu only offers what fits the selection. A greyed entry says why when you hover over it — for example, *Cut arithmetic* needs exactly two cuts, the *3D view* needs a map.

Show information also opens the full metadata of the dataset in a window of its own, with a filter box and **Export metadata (.csv)**:



Clear list asks first, and offers to save anything computed in this session that has not been saved yet. Files on disk are never deleted.

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4. Loading data

Load data... opens the loader. **Quick add...** skips it and adds the chosen files with everything detected automatically.



1. **Files** — pick one or many (*Select file(s)...*). The file filter is built from what the installed readers can open.
2. **Reader** — detected automatically, and the result is shown (*Detected: 3 × This program's own format*). Override it only if a file is being misread.
3. **Datasets found** — what is inside each file, read from its *structure* only. This is instant even for a multi-GB scan; nothing is read until you open a row.
4. **Axes** — *Axis order* reorders the array's dimensions once, at the door. *First axis of a map is* says what was scanned: a deflector angle (the default), a **photon energy**, a temperature, a gate voltage, a delay… This keeps the right unit everywhere afterwards, and stops a k conversion being offered for an axis that is not an angle.

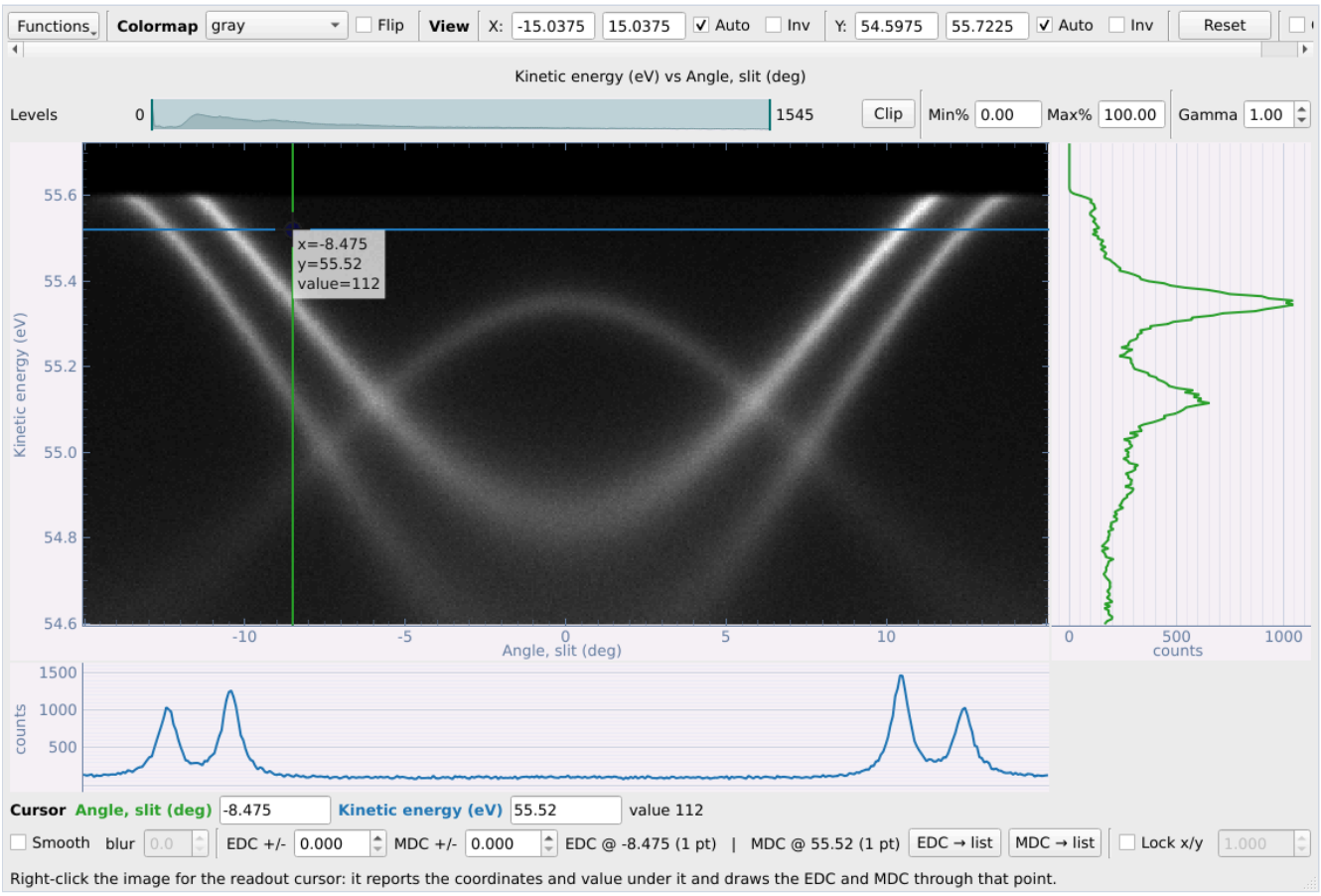
CASSIOPEE folders. Select any file of a numbered series (`name_1_ROI1_.txt`, `name_2_ROI1_.txt`, …) and the whole folder is listed once, as one assembled cube. Whether it becomes an angle map or a kz map is read from the `_i` parameter files next to the spectra.

Data kinds

Kind	Axes	What it is
`cut`	angle (or k) × E	one spectrum
`map`	angle × angle × E	a deflector or polar-angle scan
`k_map`	kx × ky × E	a map converted to momentum
`kz_map`	hν × angle × E	a photon-energy scan
`kz_map_k`	k_z × k∥ × E	that scan converted to momentum
`spem_4d`, `spem_1d`	x (× y) × angle × E	real-space scans
`edc`, `mdc`, `spin_edc`	E (or k) × channels	curves, with optional σ channels

5. What the viewers have in common

Each dataset opens in its own window, so you can put a map, its cuts and another file side by side.



- **The top row**: the **Functions** menu (everything the window can do), the colormap (per window) with **Flip**, and the view controls — axis ranges, **Auto**, **Inv** (invert), **Reset**, **Grid**.
- **Above each image**: the **Levels** bar (drag the two handles, or type **Min%/Max%** percentiles; **Clip** applies them) and **Gamma**.
- **Under each image**: **Smooth** (display only, does not change the data) and **Lock x/y** (aspect ratio).
- **Readout cursor** — right-click the image → *Readout cursor*. It reports the value under it and, on a cut, draws the **EDC** (right) and **MDC** (below) through that point. **EDC ±** / **MDC ±** integrate over a window, shown as a shaded band on the image. Type a position into the **Cursor** row and press Enter to put it exactly there.
- **EDC → list** / **MDC → list** put the curve on screen in the main list as a dataset of its own (see §15).
- **Right-click the image → Export** saves what the panel shows — data (`.nxs`), an image, or vector axes (`.eps`); see §17.
- **Functions → Slice → Save slice to the main list / Open slice in a new panel** keeps a copy of the slice on screen.

The **Functions** menu is organised in sections:

Viewer	Analysis	Data operations	Visualization
map / kz map / k-map		Arbitrary cut, Map k conversion, kz map processing, kz → momentum, De-grid map	Brillouin zone (k-maps), Slice figure

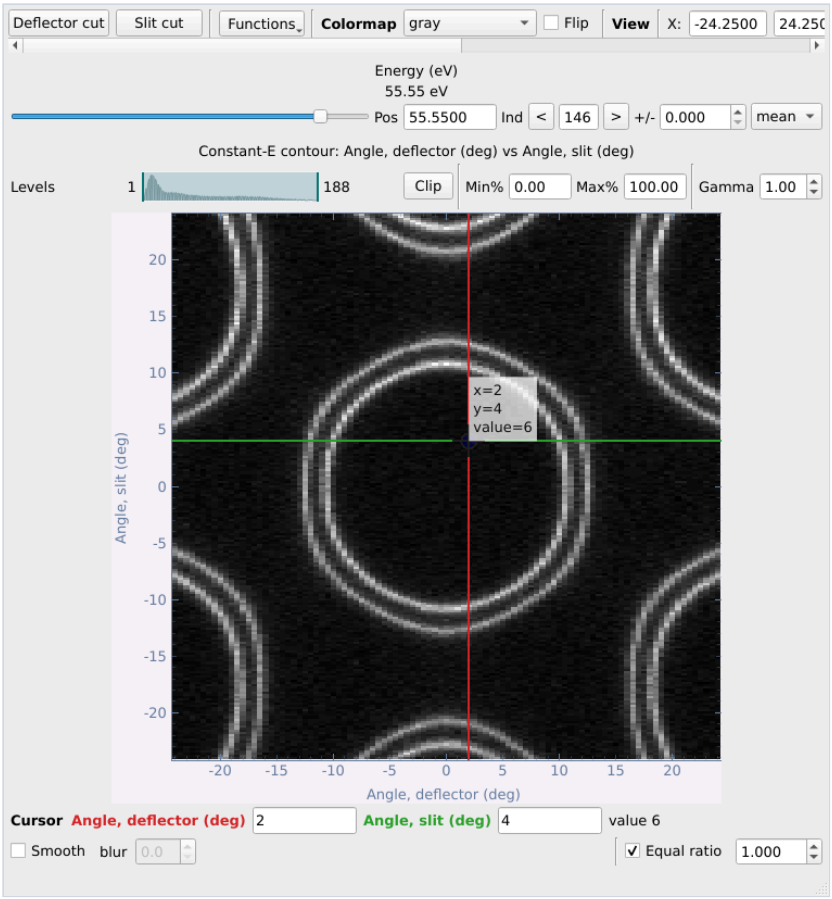
Viewer	Analysis	Data operations	Visualization
cut	Fermi level, MDC / EDC fit	FS correction, Cut k conversion, Cut arithmetic, De-grid	
curve	Curve fit, Spin analysis	Crop, Bin, Normalise, Background, Shift axis, Add √N errors	As a figure

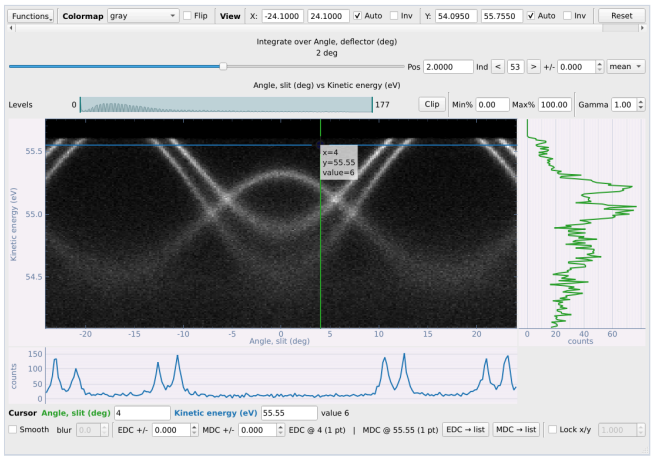
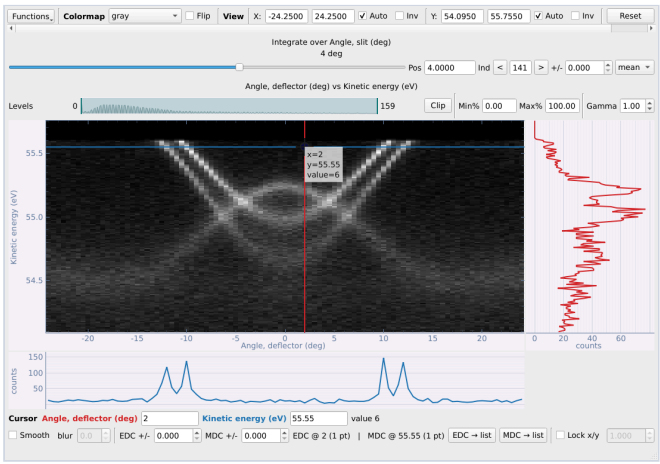
An entry that does not apply is not shown; one that applies but cannot run yet is greyed, with the reason in its tooltip.

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6. Maps: the contour and its two cuts

A map opens on its **constant-energy contour**. The slider at the top picks the energy (**Pos** in eV, or **Ind** as an index; **±** integrates over a window, **mean** or **sum**). **Deflector cut** and **Slit cut** open the two orthogonal cuts, each in its own window.



Slit cut (slit angle vs energy)	Deflector cut (deflector angle vs energy)
	

One cursor for all three windows. Switching the readout cursor on in any of them switches it on in the others, and moving it anywhere re-slices the others through the new point. The axes keep one colour each everywhere: **deflector red, slit green, energy blue**. A cursor line has the colour of the axis it holds constant, and the EDC or MDC it feeds is drawn in the same colour. Integration widths (the ± boxes) belong to each window and are never shared.

From the list you can also open a map's cut straight away (right-click → **Open slit cut** / **Open deflector cut**); the contour then stays hidden until you ask for it (*Functions → Show the map*).

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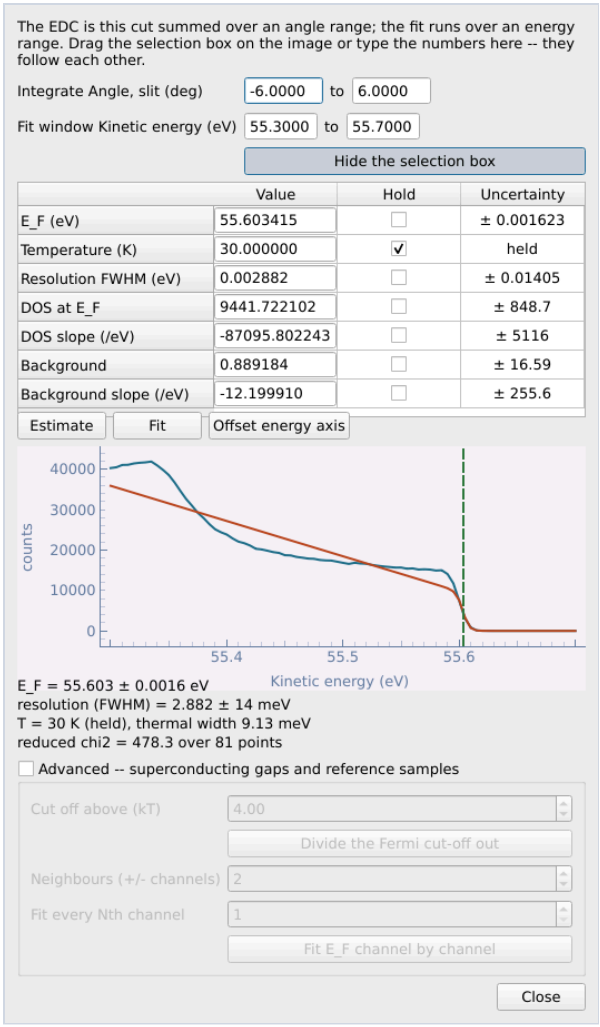
7. Cuts: from raw spectrum to band parameters

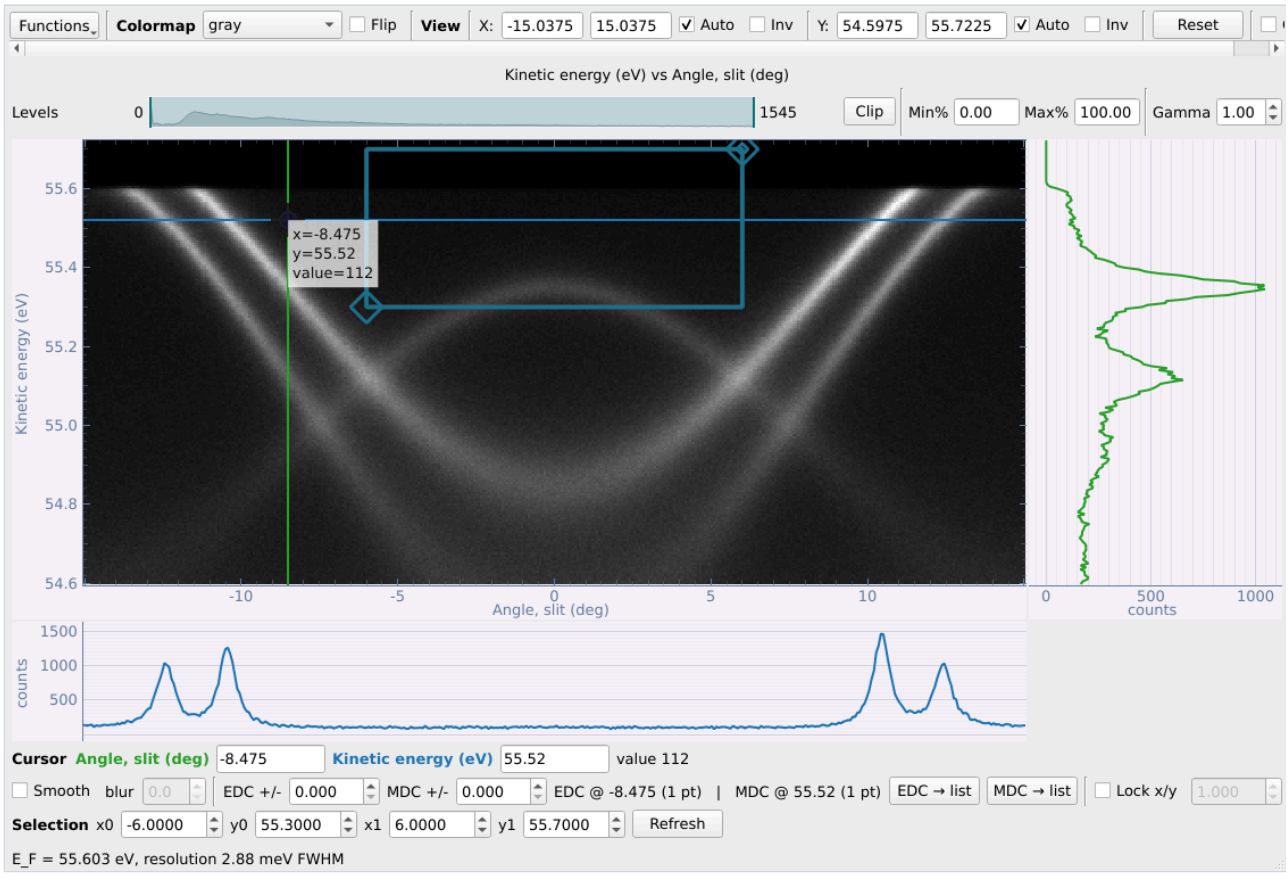
The usual order for a single cut is: **Fermi level → offset the energy axis → k conversion → MDC/EDC fit**.

7.1 Fermi level

Functions → Fermi level... fits a Fermi edge (Fermi–Dirac × linear DOS, convolved with a Gaussian resolution) to the EDC summed over a box.

1. Press **Show the selection box** and drag the box on the cut over the edge (or type the angle and energy ranges). The angle range says which channels are summed; the energy range is the fit window.





2. **Estimate**, then **Fit**. Tick **Hold** to keep a parameter fixed (the temperature is held at the recorded value by default). Every parameter is reported with its uncertainty.

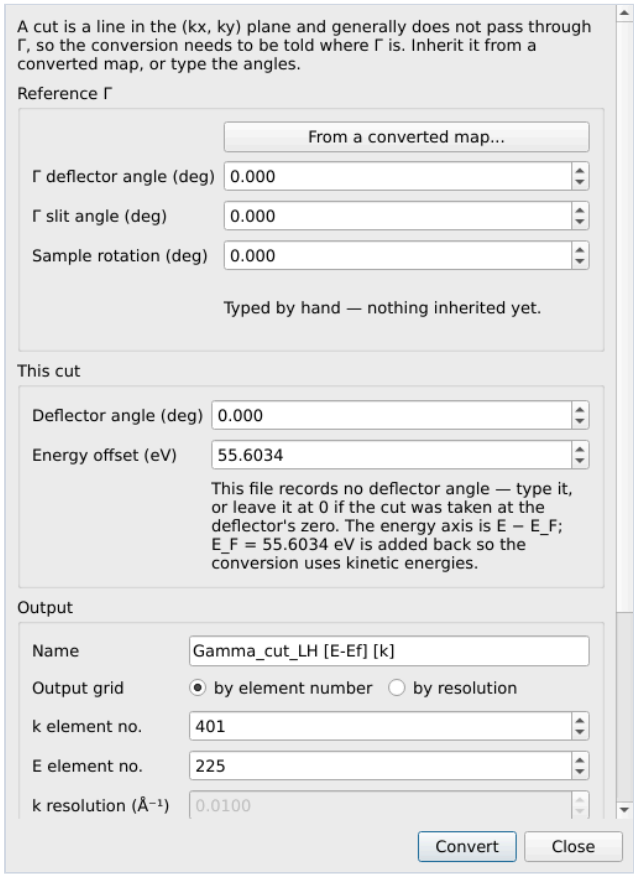
3. **Offset energy axis** lists a copy of the cut on **E − E_F**.

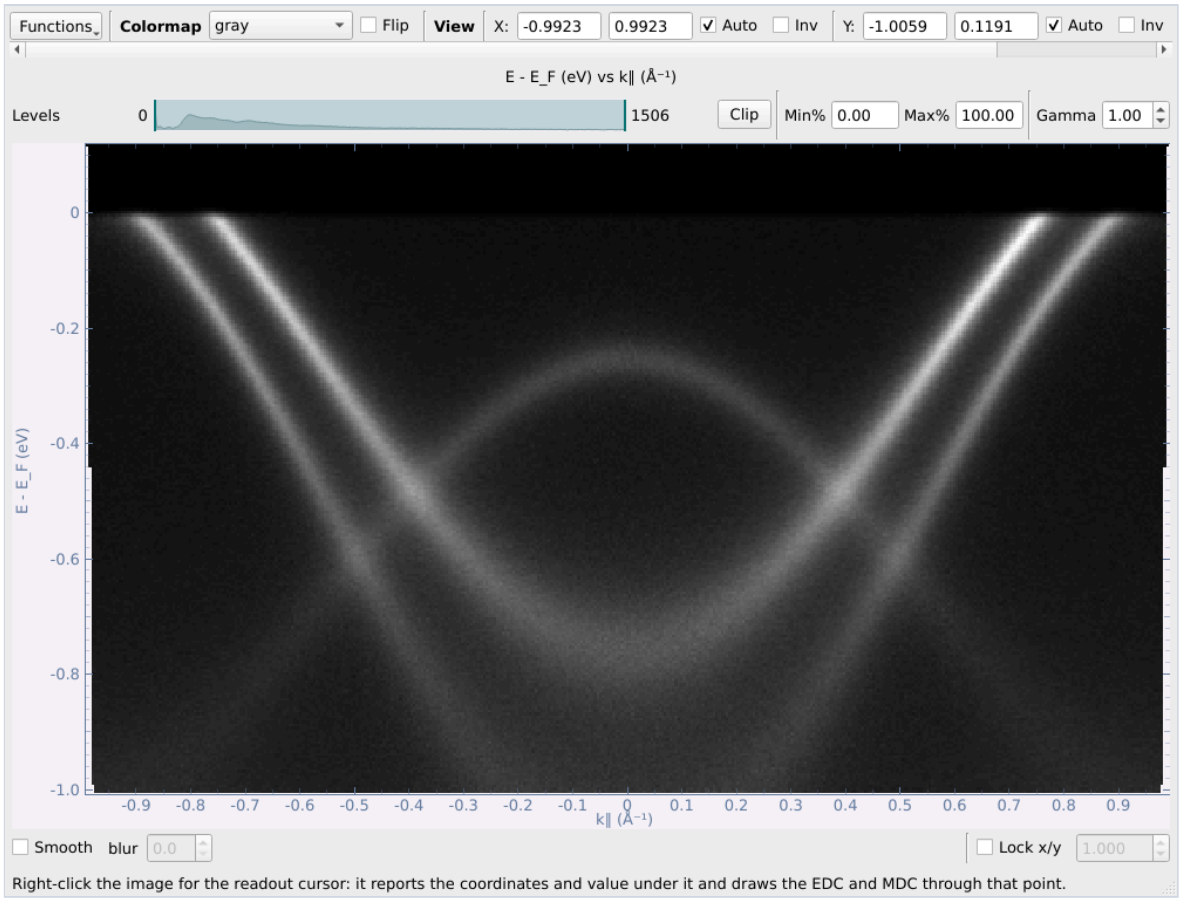
The *Advanced* corner divides the Fermi cut-off out (for gaps and reference samples) or fits E_F channel by channel.

7.2 Cut k conversion

Functions → Cut k conversion... converts the angle axis to k∥ (Å⁻¹). A cut is a line in the (kx, ky) plane that generally misses Γ, so the dialog asks where Γ is: type the angles, or **From a converted map...** to inherit them from a map already converted.

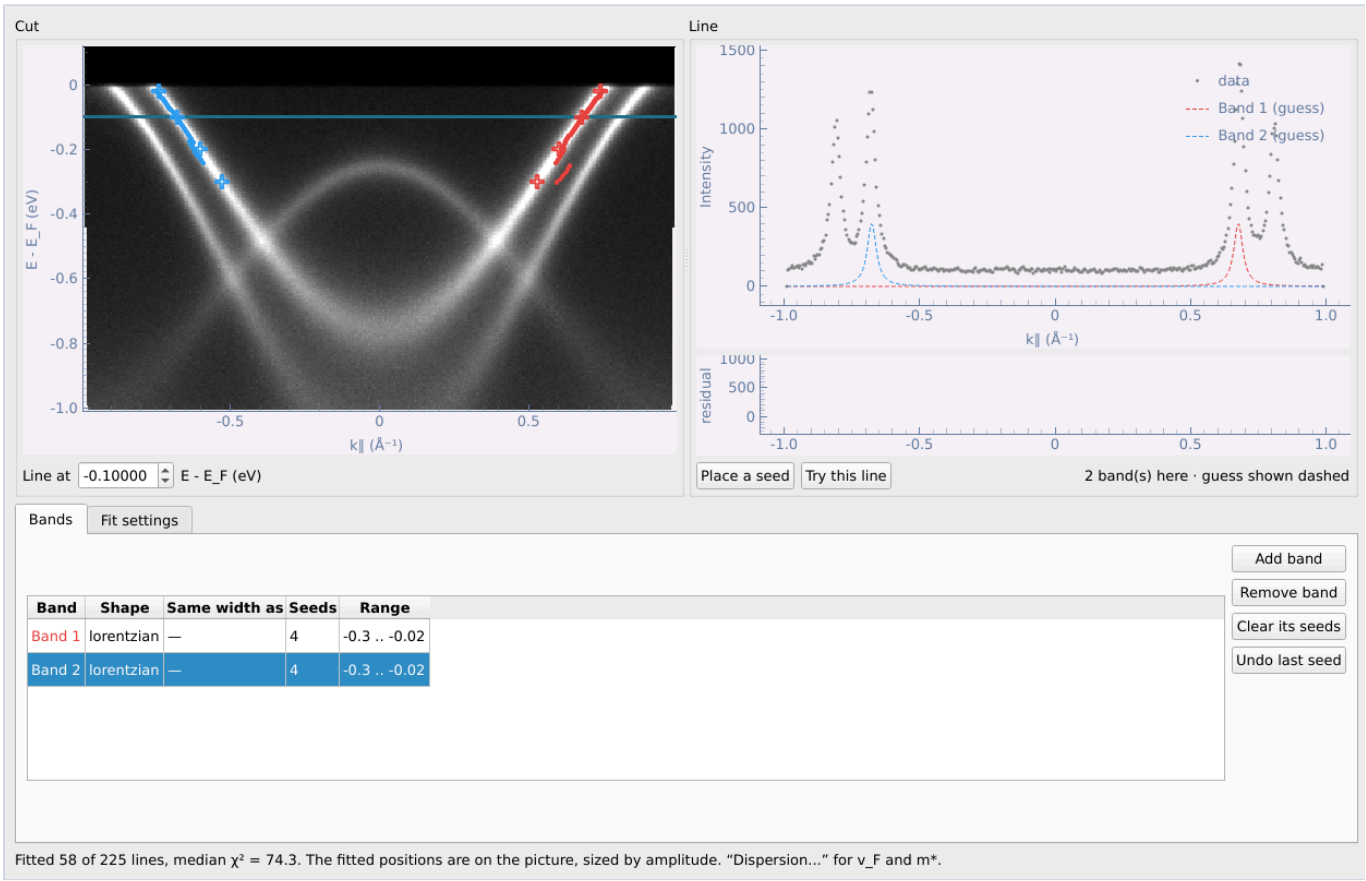
If the cut is already on E − E_F, the dialog fills in **Energy offset** with the E_F that was subtracted (the note under it says so): the conversion uses true kinetic energies, and the result stays on E − E_F.



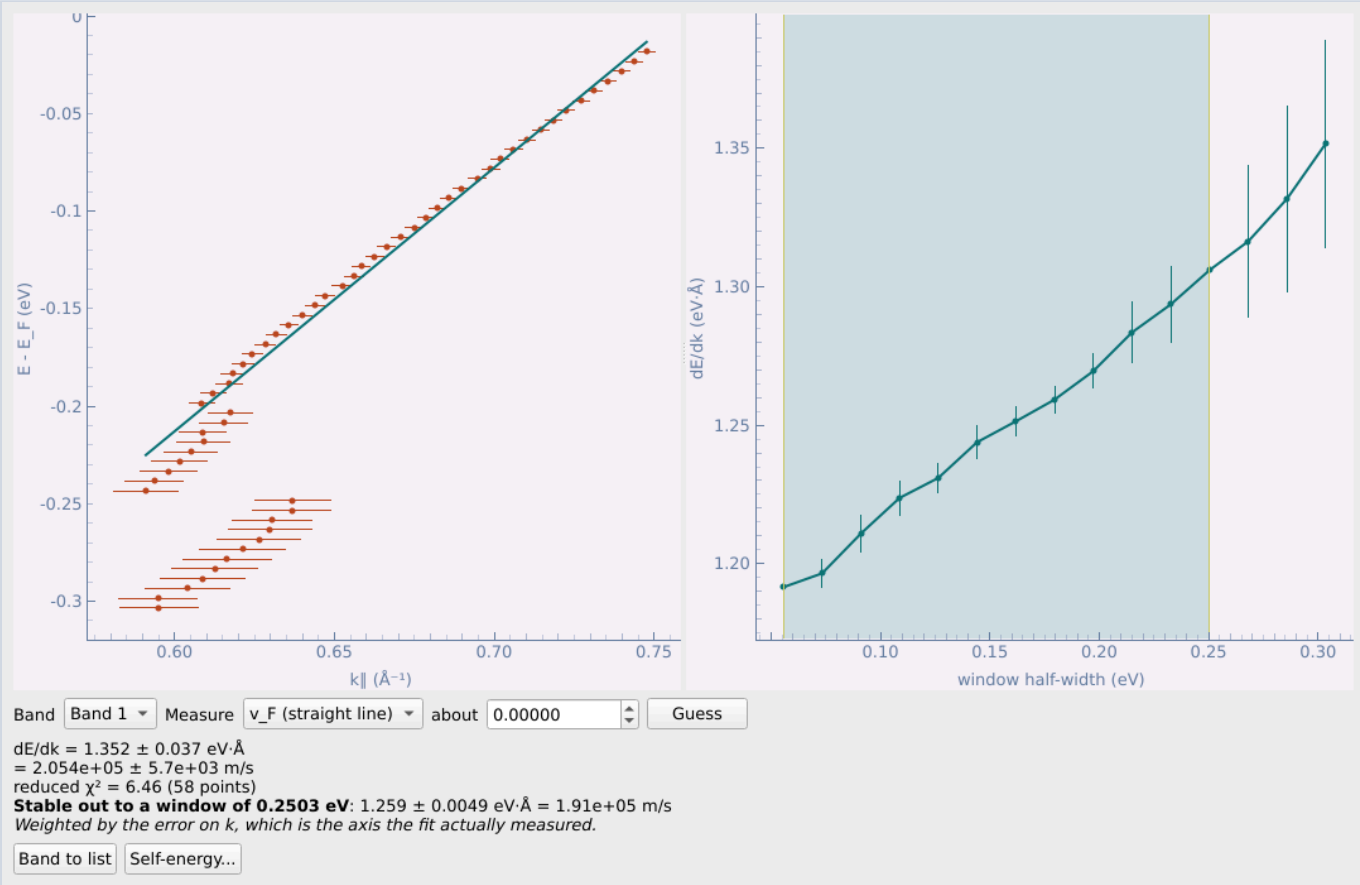


7.3 MDC / EDC fit

Functions → MDC / EDC fit... fits peaks line by line and reads the band off the fitted centres. It only accepts a cut whose axis is already in Å⁻¹ — a velocity in eV·deg would be meaningless.



1. **Bands** tab: **Add band** for each branch; select a band, press **Place a seed** and click the peak on the curve (right). Four or five seeds down a band are enough; **Line at** moves the line being looked at.
2. **Try this line** shows the fit of the current line (dashed: the guess).
3. **Fit settings** tab: MDC or EDC, peak shape (Lorentzian, Gaussian, Voigt), background, step, then **Fit the series**. Fitted positions are drawn on the image, sized by amplitude.
4. **Dispersion...** fits the band: **v_F** (straight line), **m*** (parabola) — and checks how far out the fit is stable. **Self-energy...** gives Re Σ and Im Σ.



Save table... writes every fitted line; **Band to list** puts a band in the main list.

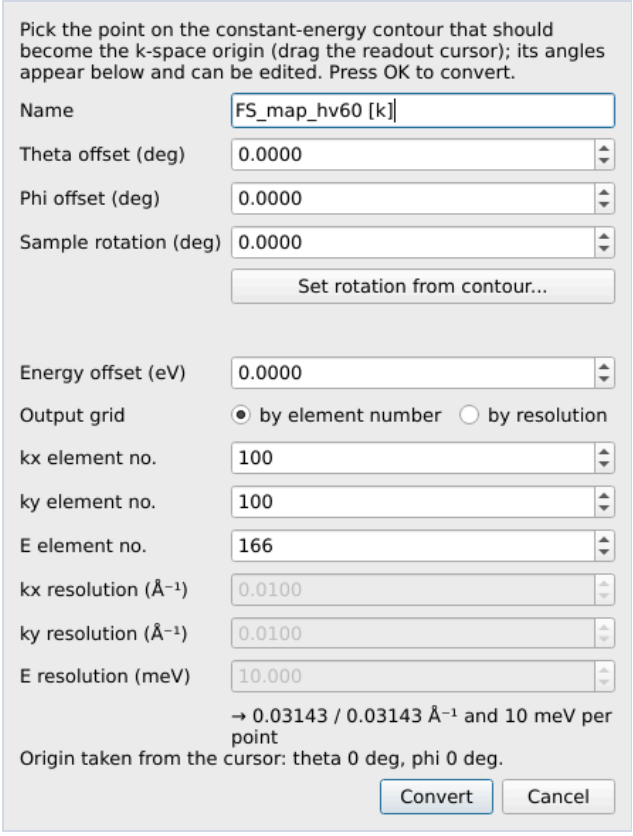
---

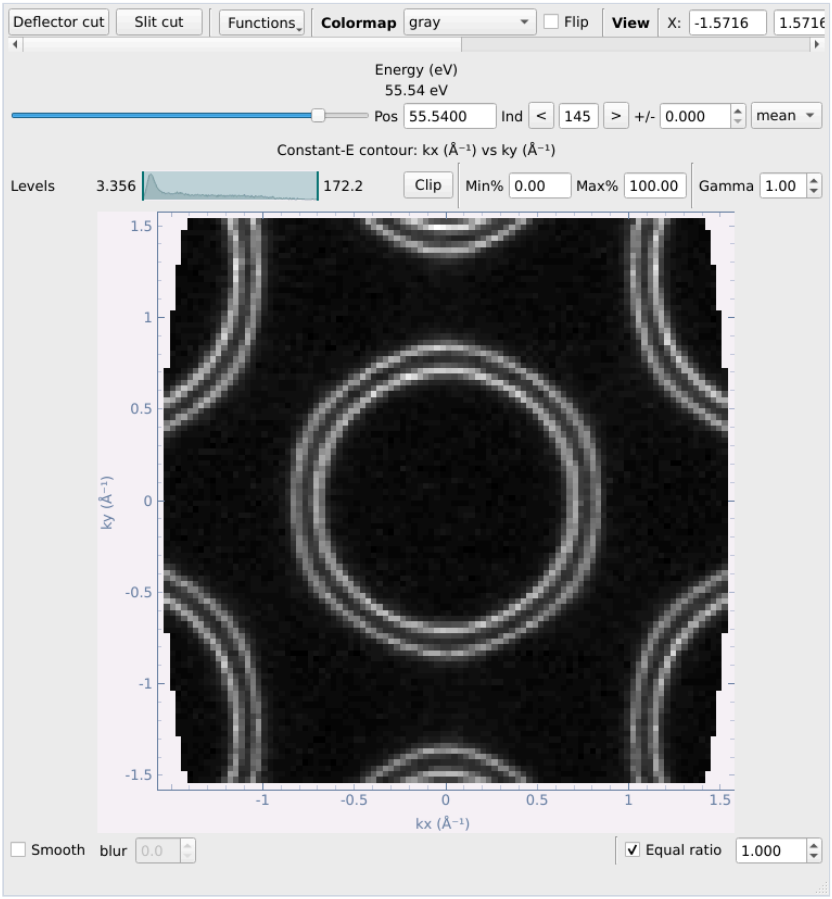
8. From angles to momentum for a map, and the Brillouin zone

8.1 Map k conversion

Functions → Map k conversion... on the contour window. Drag the readout cursor to the point that should become **k = (0, 0)** — the dialog follows it — then **Convert**. The result is a new `k-map` row in the list.

- **Sample rotation** turns the map about the surface normal; **Set rotation from contour...** reads it off a direction you click on the contour (two points along a mirror line, say).
- **Output grid**: number of points, or resolution (Å⁻¹, meV).
- **Energy offset** is added to the energy axis before converting, for an axis that is not a true kinetic energy (filled in automatically for a map on E − E_F).

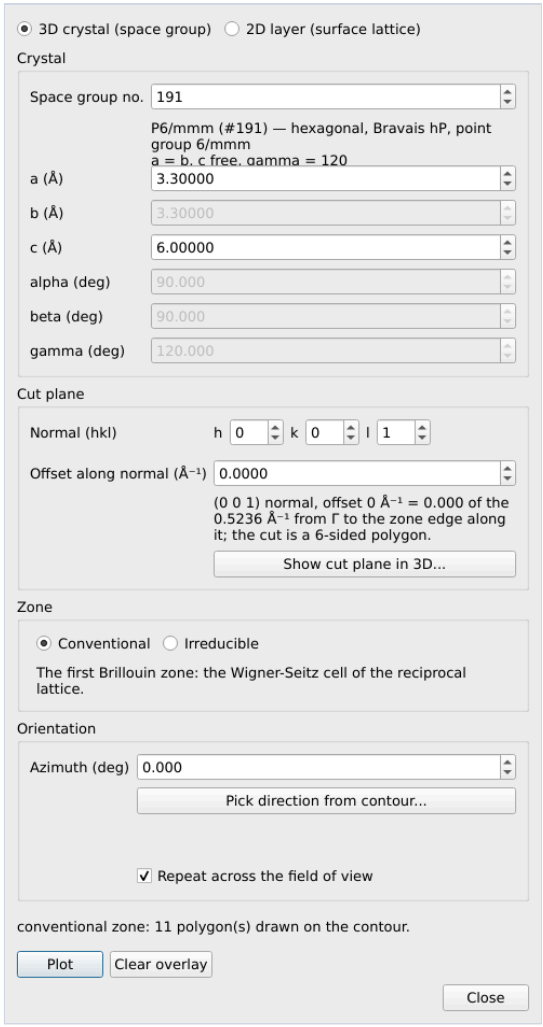


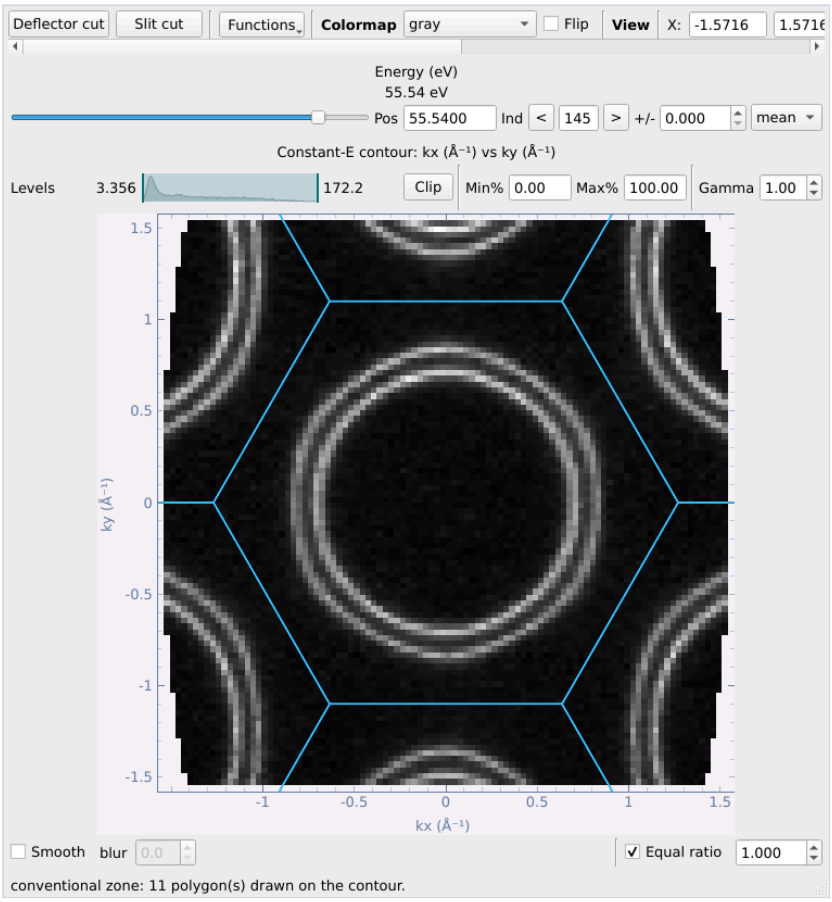


8.2 Brillouin zone

On a k-map: *Functions → Visualization → Brillouin zone...*.

1. **Space group** first (1–230); the lattice parameters it fixes are greyed and follow the free ones. Or switch to **2D layer** for a surface lattice.
2. **Cut plane**: the surface normal (hkl) and an offset along it — the cut of the 3-D zone by that plane is drawn, for any surface, not only (001). **Show cut plane in 3D...** previews it.
3. **Conventional** or **Irreducible** zone; **Azimuth** (or **Pick direction from contour...**); **Repeat across the field of view**.
4. **Moiré** adds a second, twisted or mismatched layer and its moiré zone.
5. **Plot**.

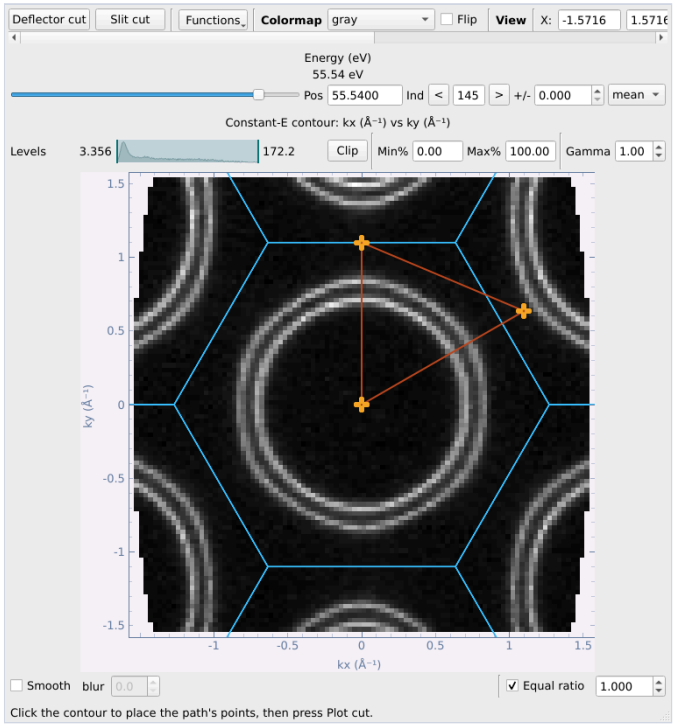
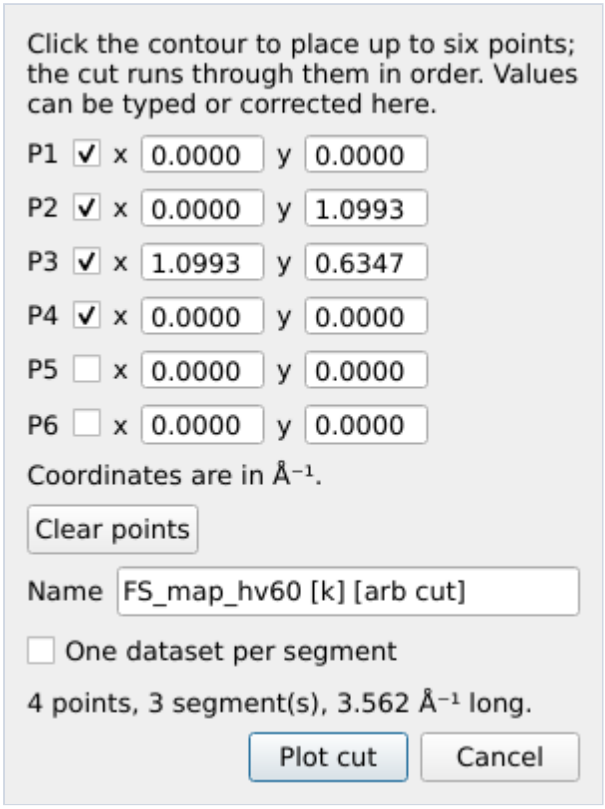




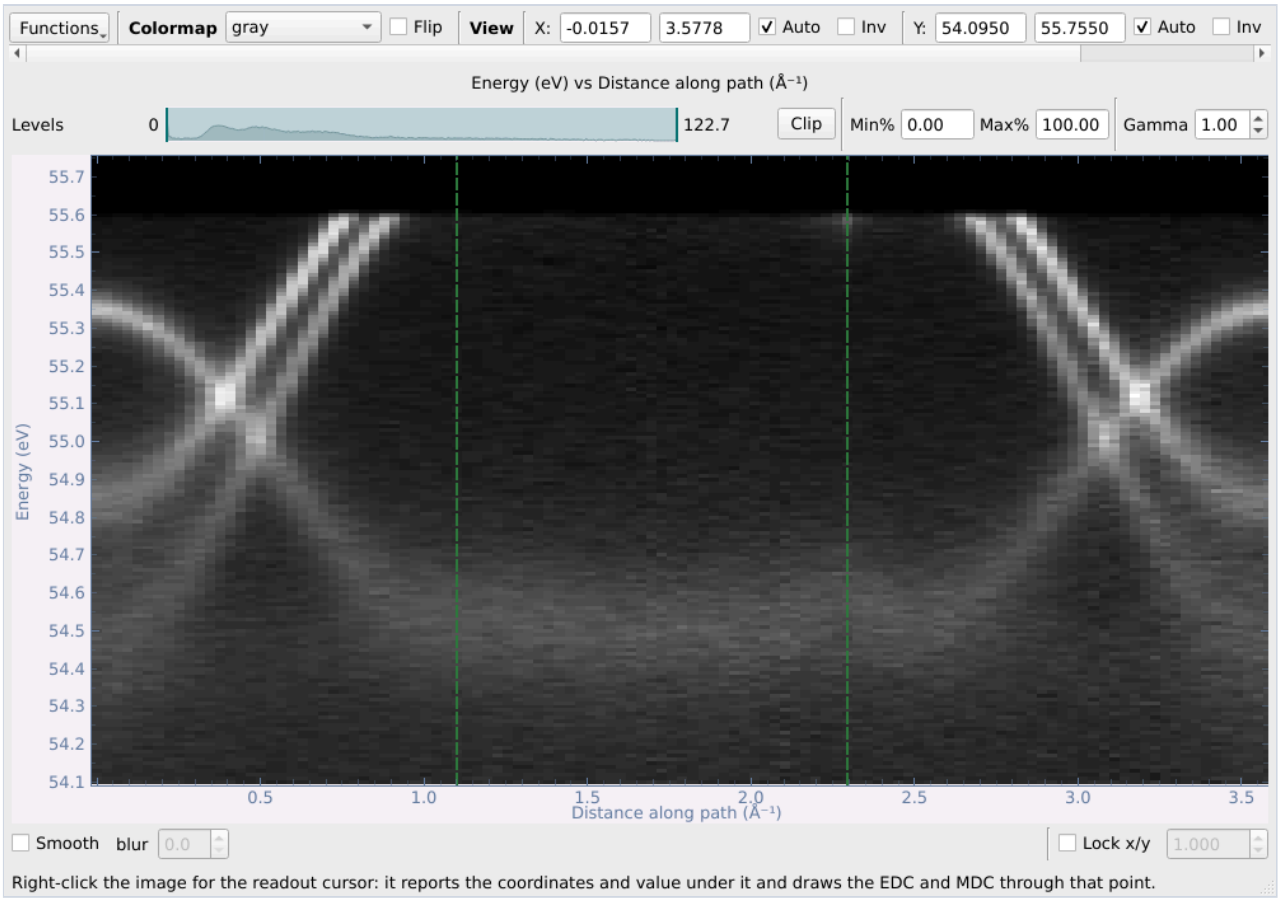
9. Arbitrary cuts and Fermi-surface correction

9.1 Arbitrary cut

Functions → Arbitrary cut… on any contour: click up to six points (or type them) and the cut runs through them in order — here Γ → M → K → Γ on the k-map. **One dataset per segment** lists each leg separately.

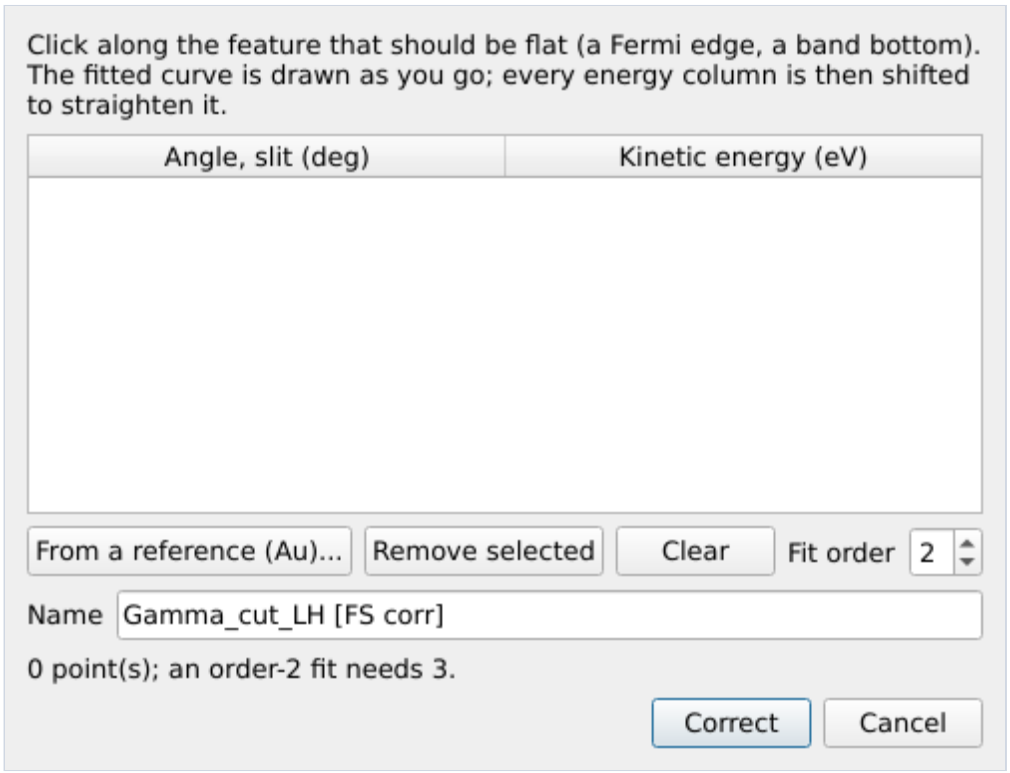
Path on the contour	Points
	

The result is a cut against the distance along the path; the corners of the path are marked with dashed green lines, so a kink there is not mistaken for physics.



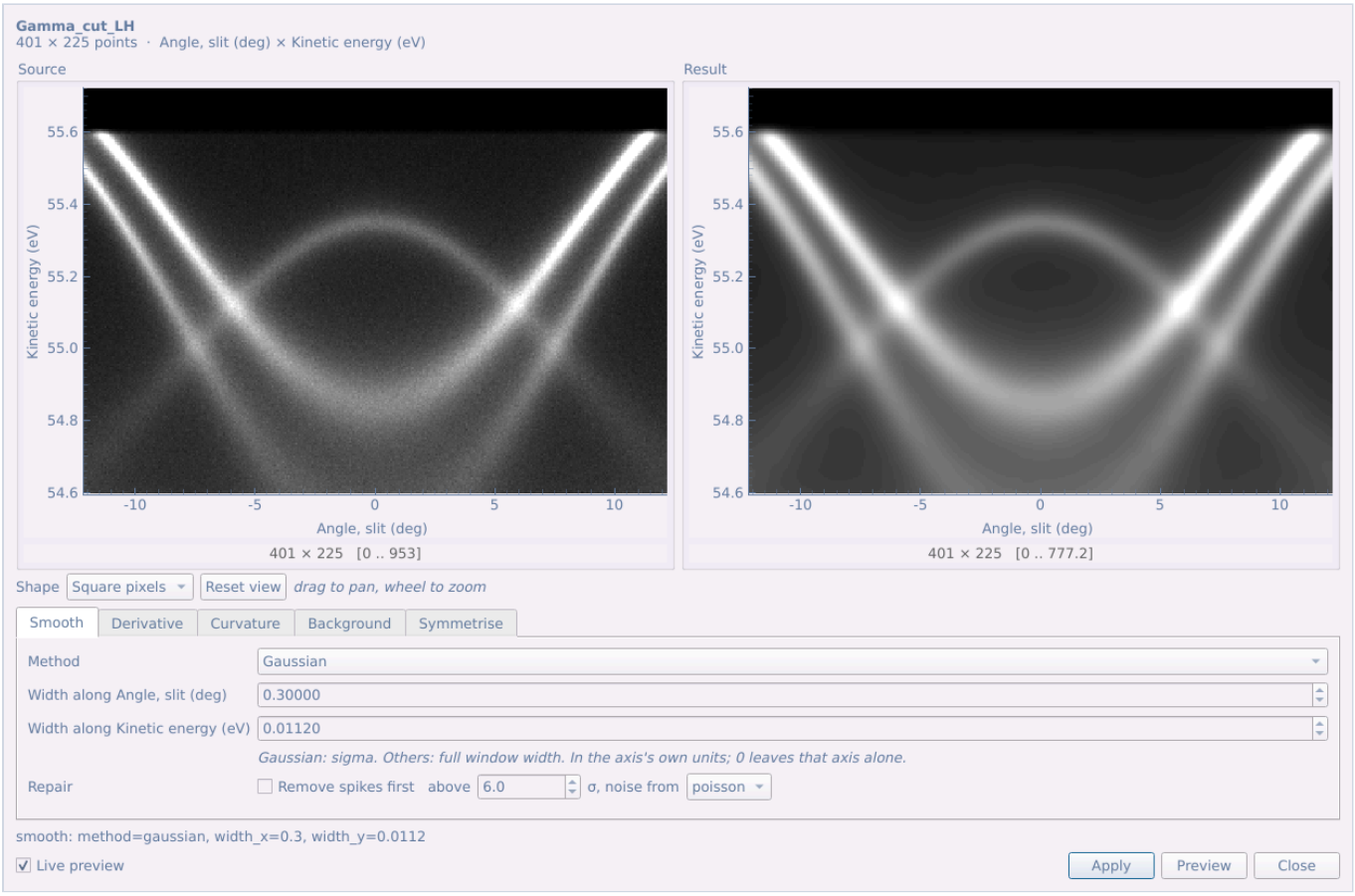
9.2 Fermi-surface (FS) correction

Functions → FS correction... on a cut (or a map's slit cut) straightens a feature that should be flat — usually the Fermi edge, which curves across the analyser angle. Click along the feature; a polynomial (order 2 by default) is fitted and every angle column is shifted in energy. **From a reference (Au)...** takes the curve from a gold spectrum in the list instead. On a map's slit cut the same shift is applied to the whole cube.



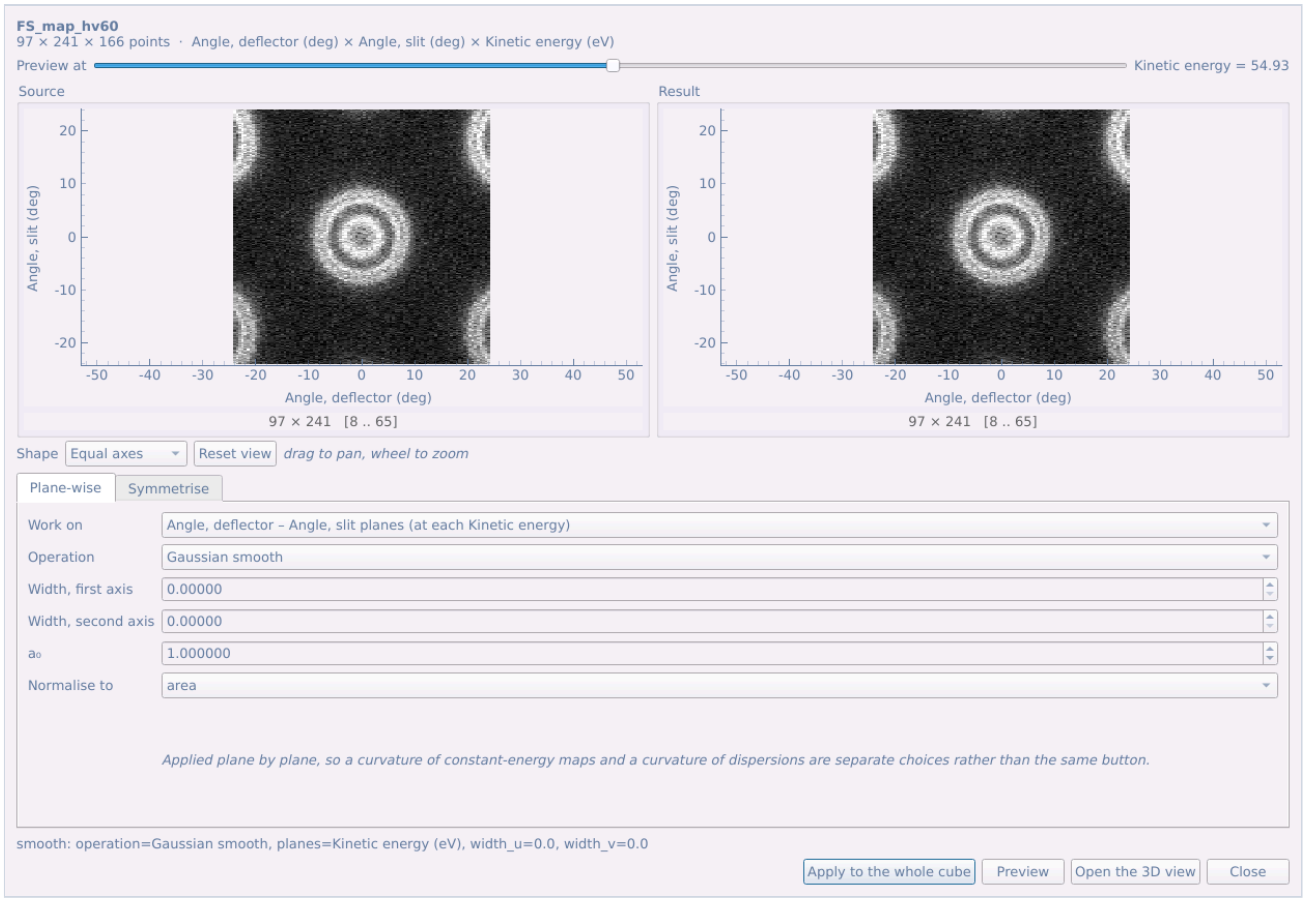
10. Processing: smoothing, derivatives, curvature

Select one or more cuts (or maps) in the list and press **Process…**. Every result is a new dataset carrying the steps that made it.

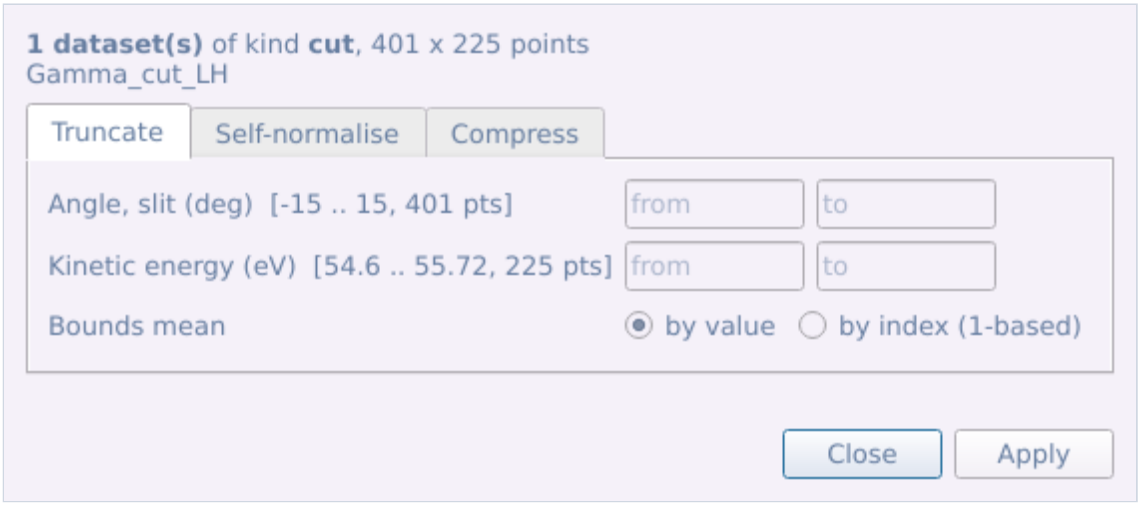


- Tabs: **Smooth** (Gaussian, box, …; optional despiking), **Derivative**, **Curvature** (1-D and 2-D), **Background**, **Symmetrise**.
- **Source** and **Result** are shown side by side on the same scale, so you can see when a setting has eaten the band. They pan and zoom together.
- **Shape**: *Square pixels* (one data point is square — undistorted for an angle-vs-energy cut), *Equal axes* (one unit is the same length on both axes — right for angle-vs-angle or k-vs-k; chosen automatically when both axes have the same unit), or *Fill the box*.
- **Live preview** re-computes as you change a value; **Apply** lists the result.

For a map the panel works plane by plane, and asks **which planes** first — constant-energy maps or dispersions — since the curvature of a contour and the curvature of a cut are different quantities. **Preview at** picks the plane shown; **Apply to the whole cube** runs it on every plane.

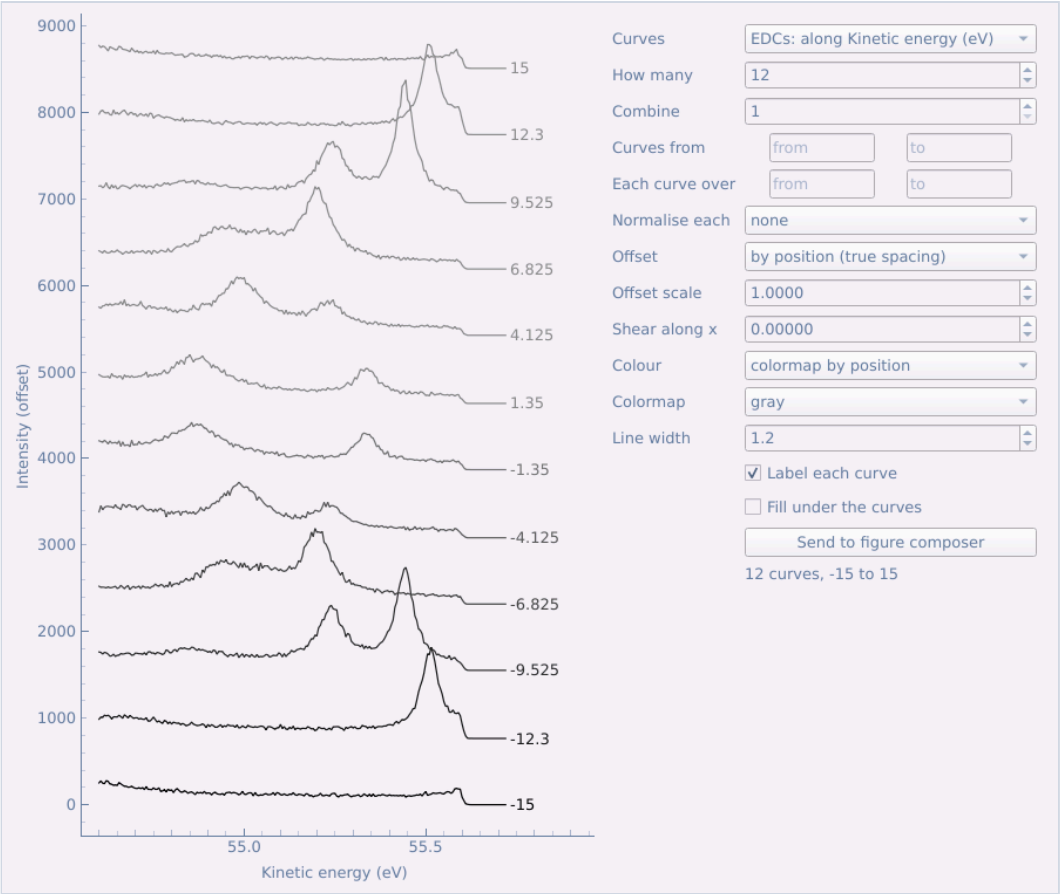


Data operations… (main window) truncates, self-normalises or compresses (bins) one or several datasets at once — as long as they have the same kind and the same number of points on every axis.

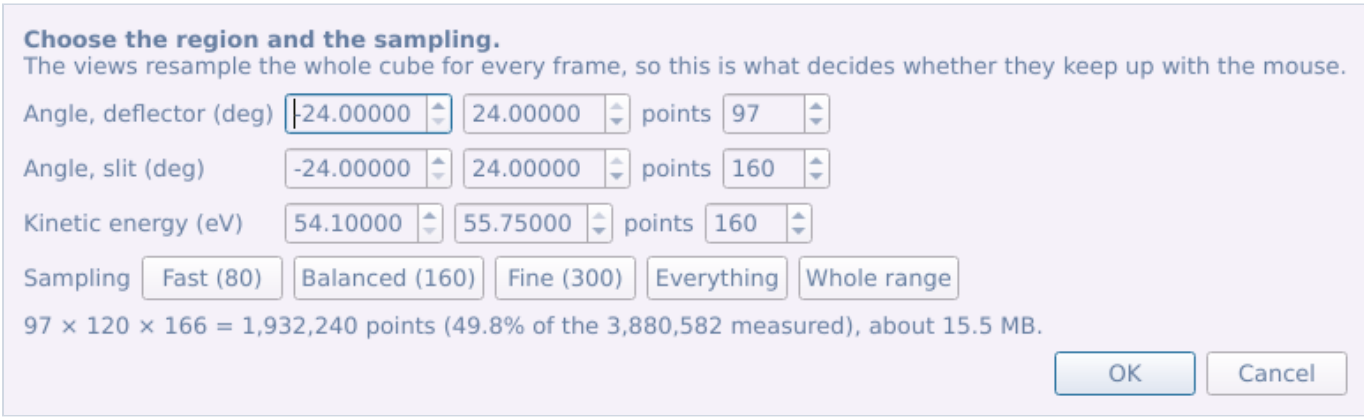


11. Stack plots and the 3-D view

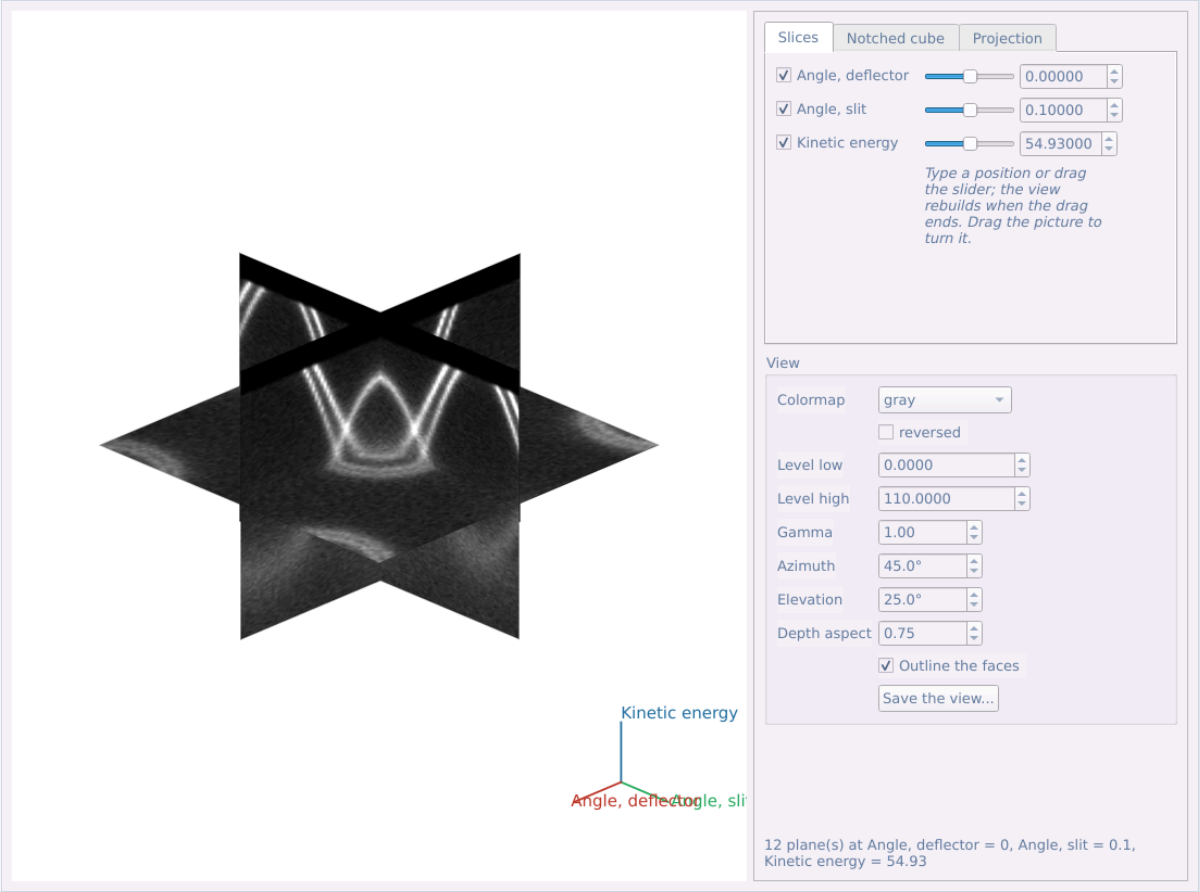
Stack plot (right-click a cut): a waterfall of EDCs or MDCs, with the number of curves, binning, normalisation, offset, shear and colouring controls; **Send to figure composer** makes it a figure panel.



3D view (right-click a map) first asks for the region and the sampling, since every frame resamples the cube — *Fast*, *Balanced* or *Fine* decide whether it keeps up with the mouse.

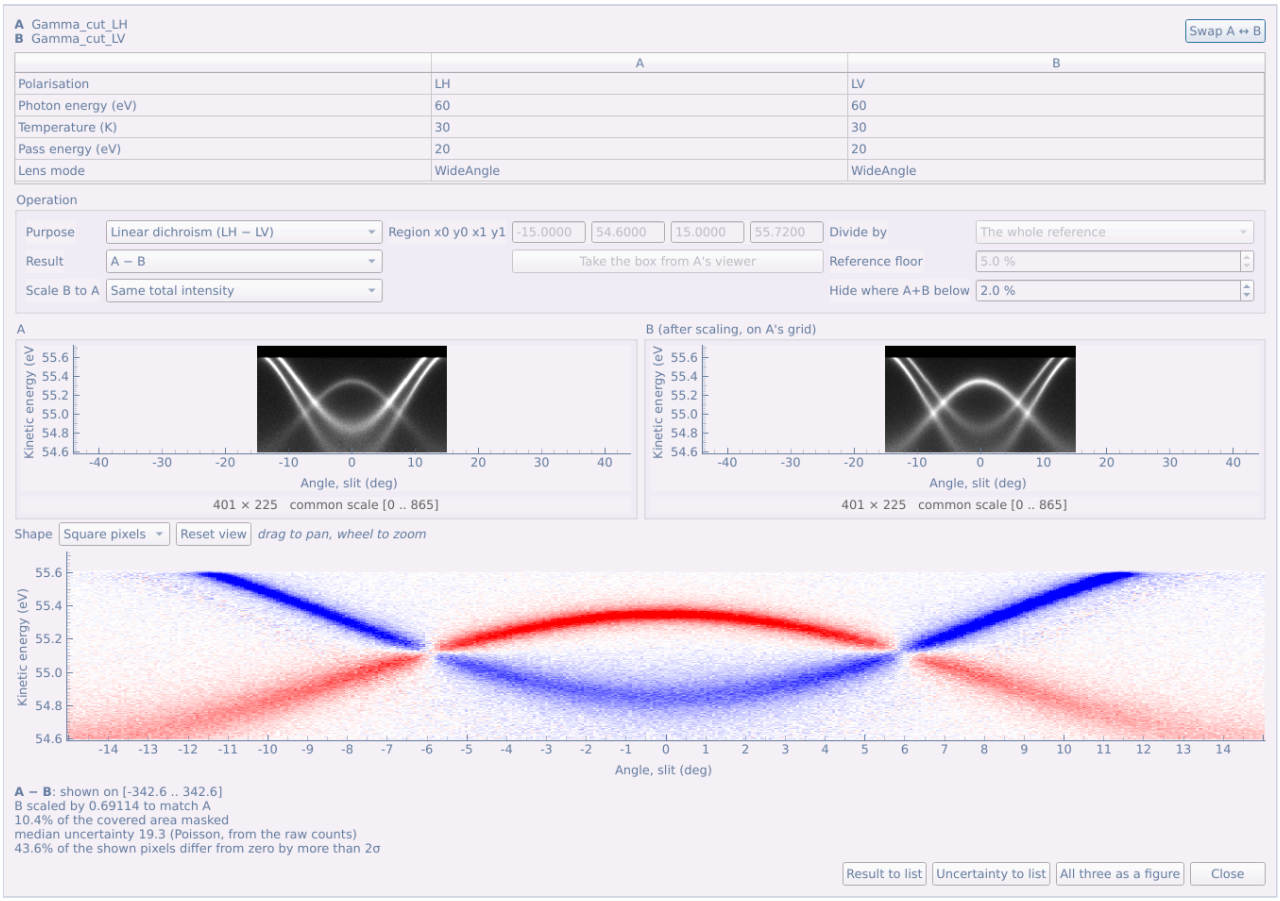


Then drag the picture to turn it. **Slices** shows three orthogonal planes at the positions you set, **Notched cube** cuts a corner out of the cube, and **Projection** sums along an axis. Colormap, levels, gamma, azimuth, elevation and depth aspect are on the right; **Save the view...** exports the picture.



12. Cut arithmetic: dichroism and reference division

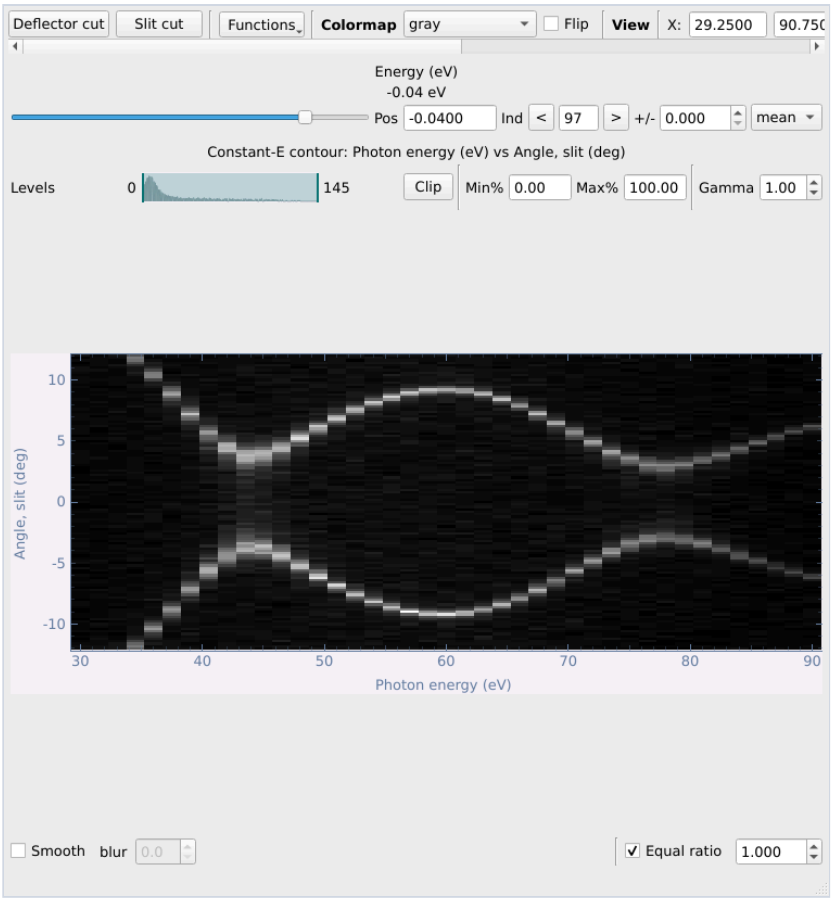
Two cuts combined into a third: linear dichroism (LH − LV), circular dichroism, dividing by a gold reference, or plain differences, ratios and sums. Select two cuts in the list → right-click → **Cut arithmetic on the two...** (the first selected is A), or in a cut's viewer *Functions* → *Cut arithmetic...* and then click the other cut in the list.



- **The table at the top** compares what the two measurements recorded (polarisation, photon energy, angles, sample position, temperature, pass energy, lens mode). Rows in orange differ beyond a tolerance — a dichroism map between two spots on the sample is a map of *that* difference.
- **Purpose** presets: *Linear dichroism* (A − B, B scaled to the same total intensity), *Circular dichroism* ((A − B)/(A + B)), *Divide by a reference*, *Custom*. Warnings appear if the polarisations look swapped (**Swap A ↔ B**).
- **Scale B to A** by total intensity or by a region; **Hide where A+B below** blanks the background.
- A and B are drawn on one colour scale; the result on a red-white-blue scale symmetric about zero. With raw counts on both, the Poisson uncertainty is propagated and summarised under the plot.
- **Result to list**, **Uncertainty to list**, **All three as a figure**.

13. Photon-energy scans (kz)

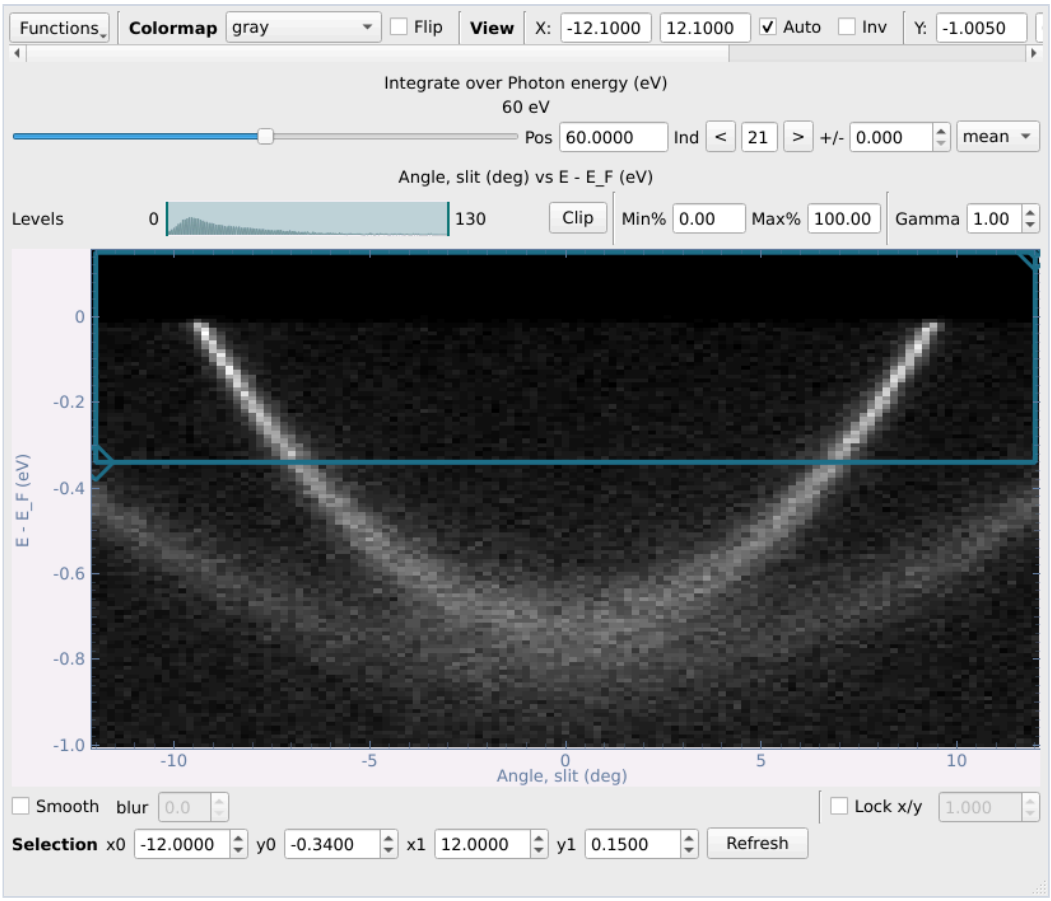
A photon-energy scan is listed as a `kz map`; its contour is photon energy against angle at one energy.



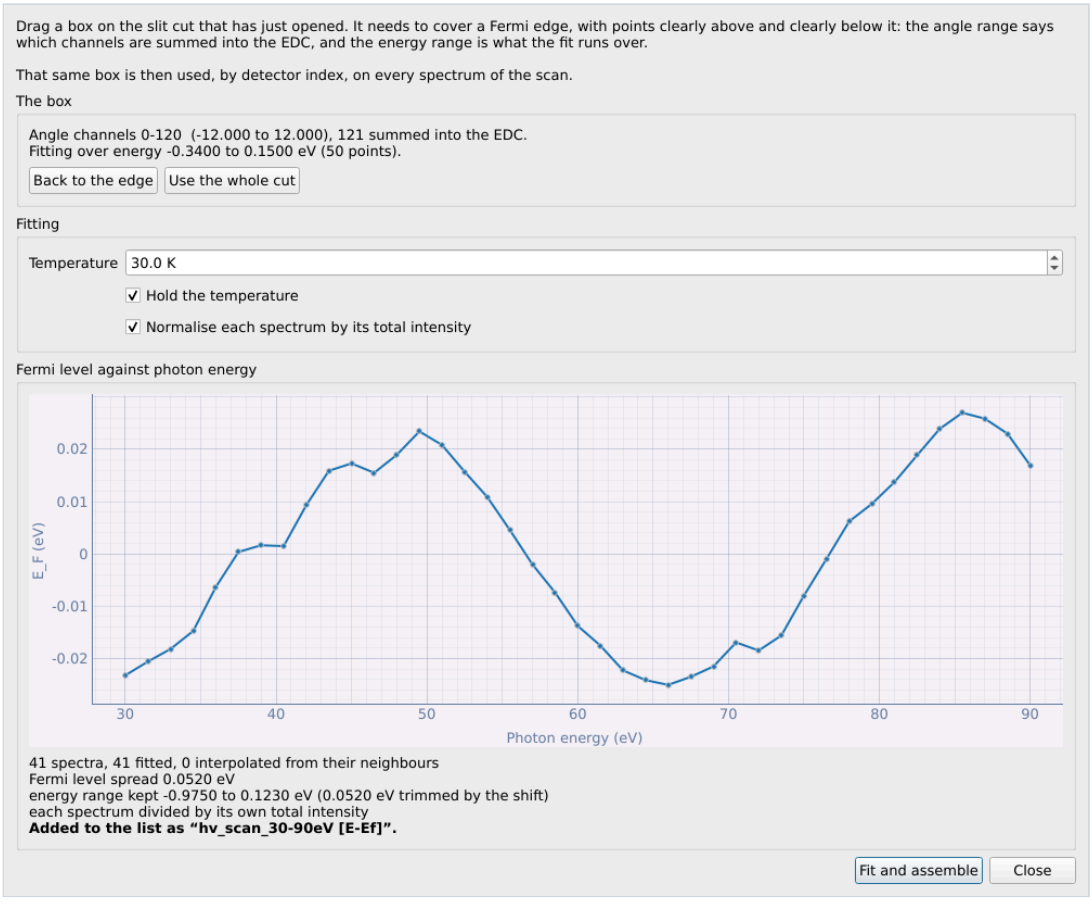
13.1 kz map processing

Nothing in the files says where each spectrum's Fermi level really is. *Functions → kz map processing...* measures it from the spectra themselves.

1. The slit cut opens with a box on it. Drag the box over a Fermi edge — the angle range sets the channels summed, the energy range the fit window.



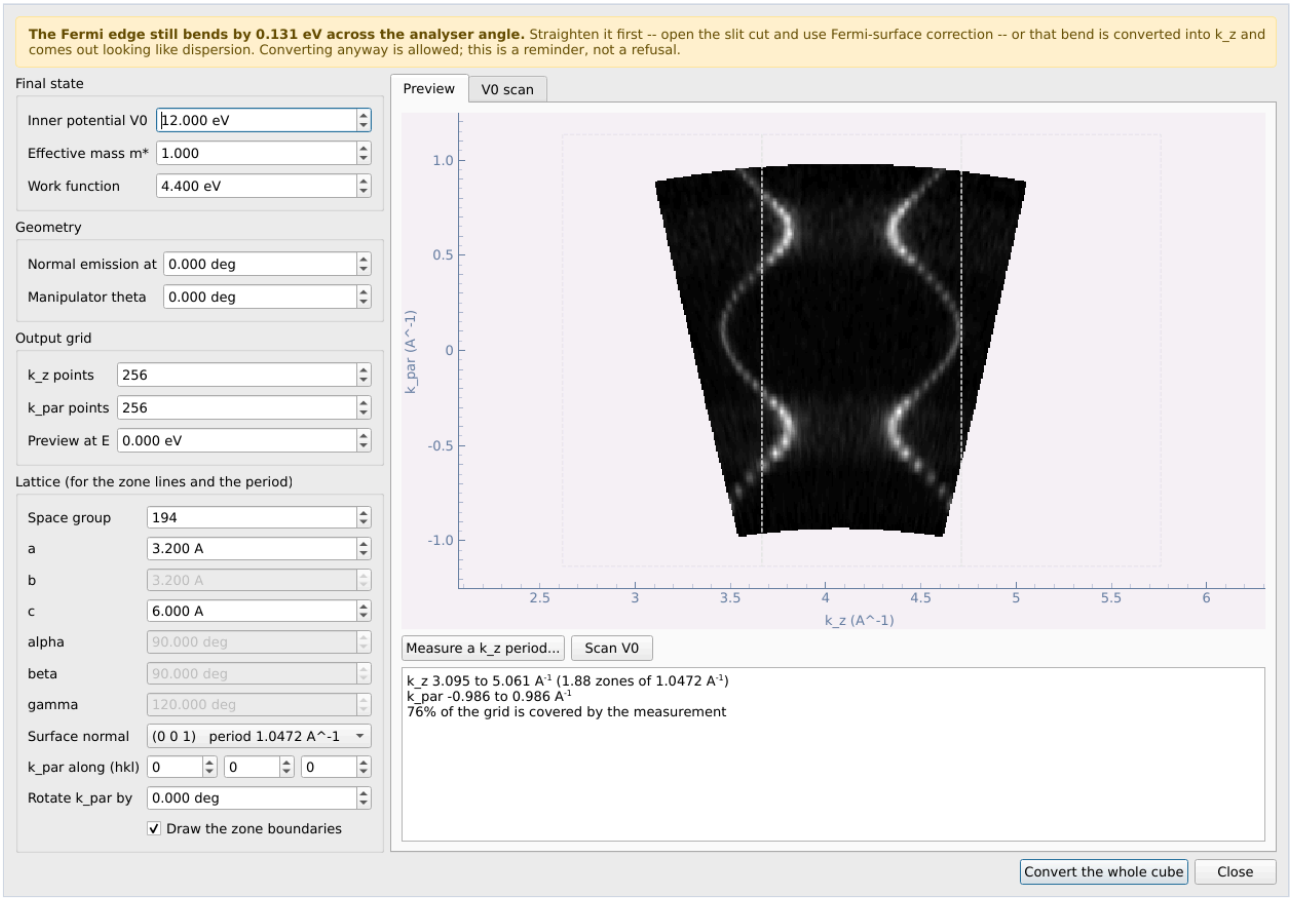
2. **Fit and assemble**: the same box (by detector index) is used for every spectrum; each is shifted to put its own edge at zero; the cube is cropped to the range they all cover; and, by default, each spectrum is **normalised by its total intensity**, which divides out the photon flux.



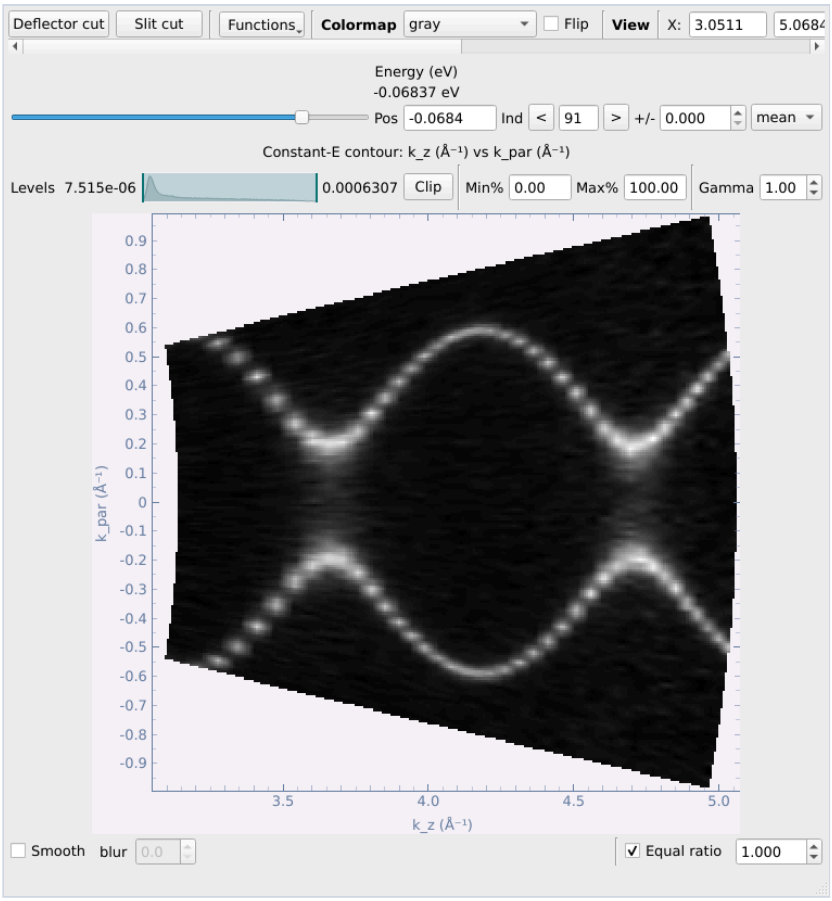
E_F against photon energy is plotted. A smooth curve means the fits are sound; points whose fit is not believed are marked and filled in from their neighbours.

13.2 kz → momentum

Functions → kz -> momentum... converts the scan to (k_z, k∥).



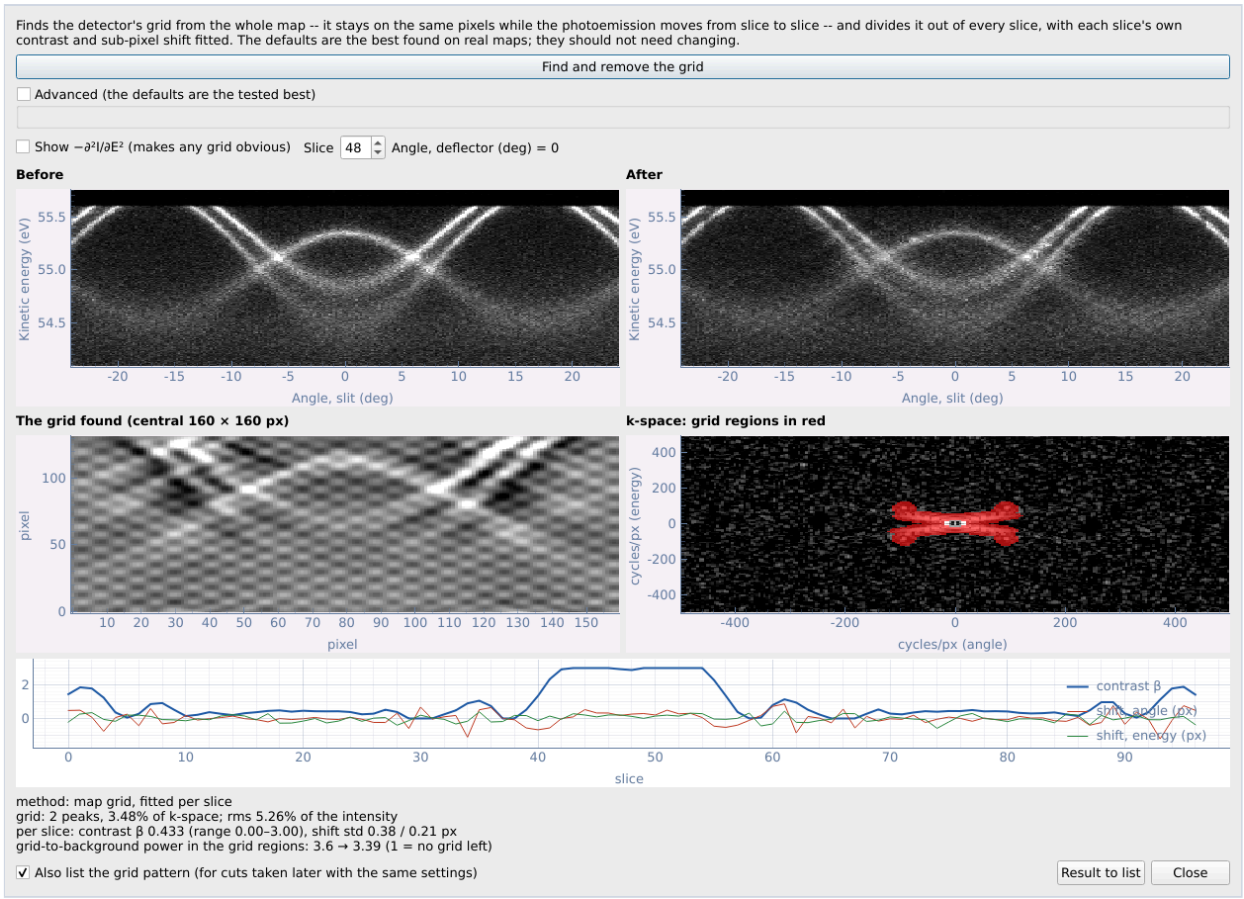
- **Inner potential V0** is the unknown: the preview reconverts one energy slice as you change it, in milliseconds.
- With the **lattice** (space group, a, b, c), the **surface normal** and the in-plane direction, the zone boundaries are drawn over the preview.
- **Scan V0** converts at a range of V0 values and reports where the k_z period matches the lattice — and how well that pins V0.
- **Measure a k_z period...**: click two points that are the same feature one zone apart; the matching reciprocal-lattice planes are listed.
- The yellow banner warns when the Fermi edge still bends across the analyser angle — correct it first (FS correction on the slit cut).
- **Convert the whole cube** lists a `kz map (k)`.



14. Removing the detector grid

MCP (hexagonal) and mesh (square) patterns print on every image. **Do it first**: only data still on the detector's pixels can be de-gridded, so the button refuses anything already converted, corrected or smoothed, and says why.

Functions → De-grid map... on a map (or its slit cut), or *De-grid...* on a cut. For a map, no reference is needed — the grid stays on the same pixels in every slice while the photoemission moves. **Find and remove the grid**:

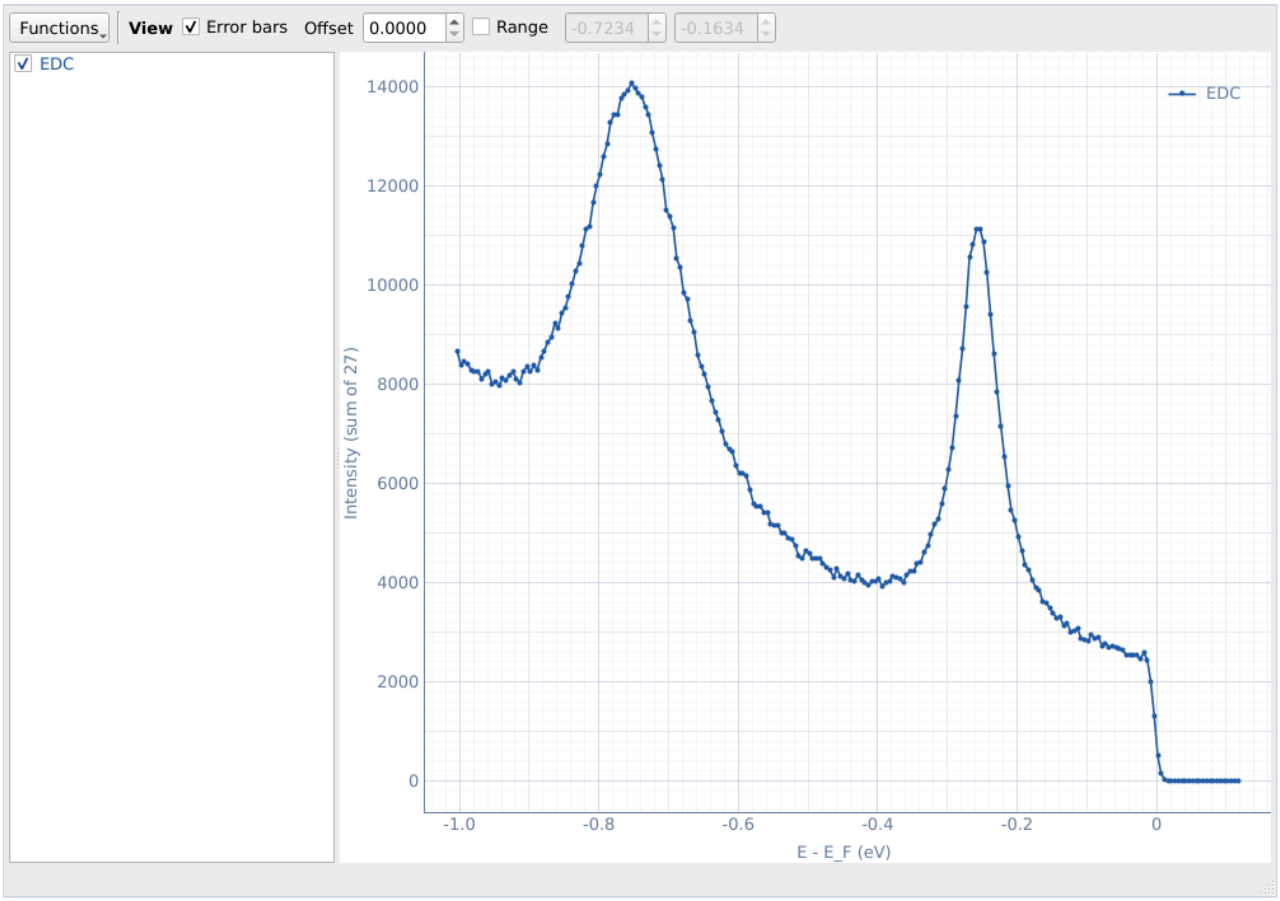


- **Before / After** a chosen slice on one colour scale (tick **Show −∂²I/∂E²** to make any grid obvious).
- **The grid found**, its **k-space regions** (in red), and the **contrast and shift** fitted for every slice.
- The power left in the grid regions (1 = no grid left).
- **Also list the grid pattern** keeps a `[grid]` dataset, which a cut taken later with the same lens mode and pass energy will find and use.

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15. Curves: EDCs, MDCs and spin-resolved data

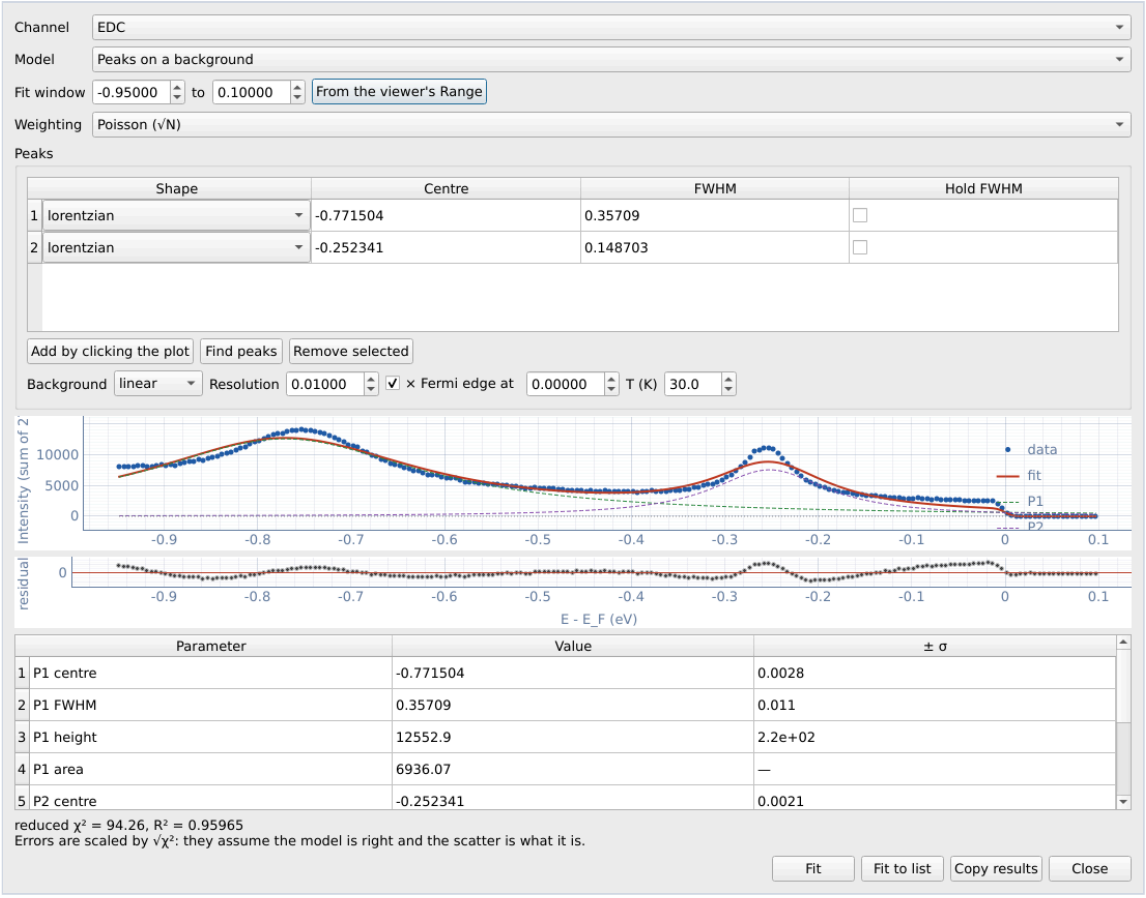
EDC → list / **MDC → list** under any image, or a spin-resolved file, give a curve dataset, which opens in the **curve viewer**:



Channels can be hidden, offset into a waterfall (**Offset**), shown with **Error bars** (from a σ channel), and **Range** marks an x window used by the operations. *Functions → Data operations* gives **Crop**, **Bin**, **Normalise**, **Subtract a background** (constant, linear, Shirley), **Shift the axis** and **Add counting errors (√N)**; each makes a new dataset and propagates σ correctly.

15.1 Curve fit

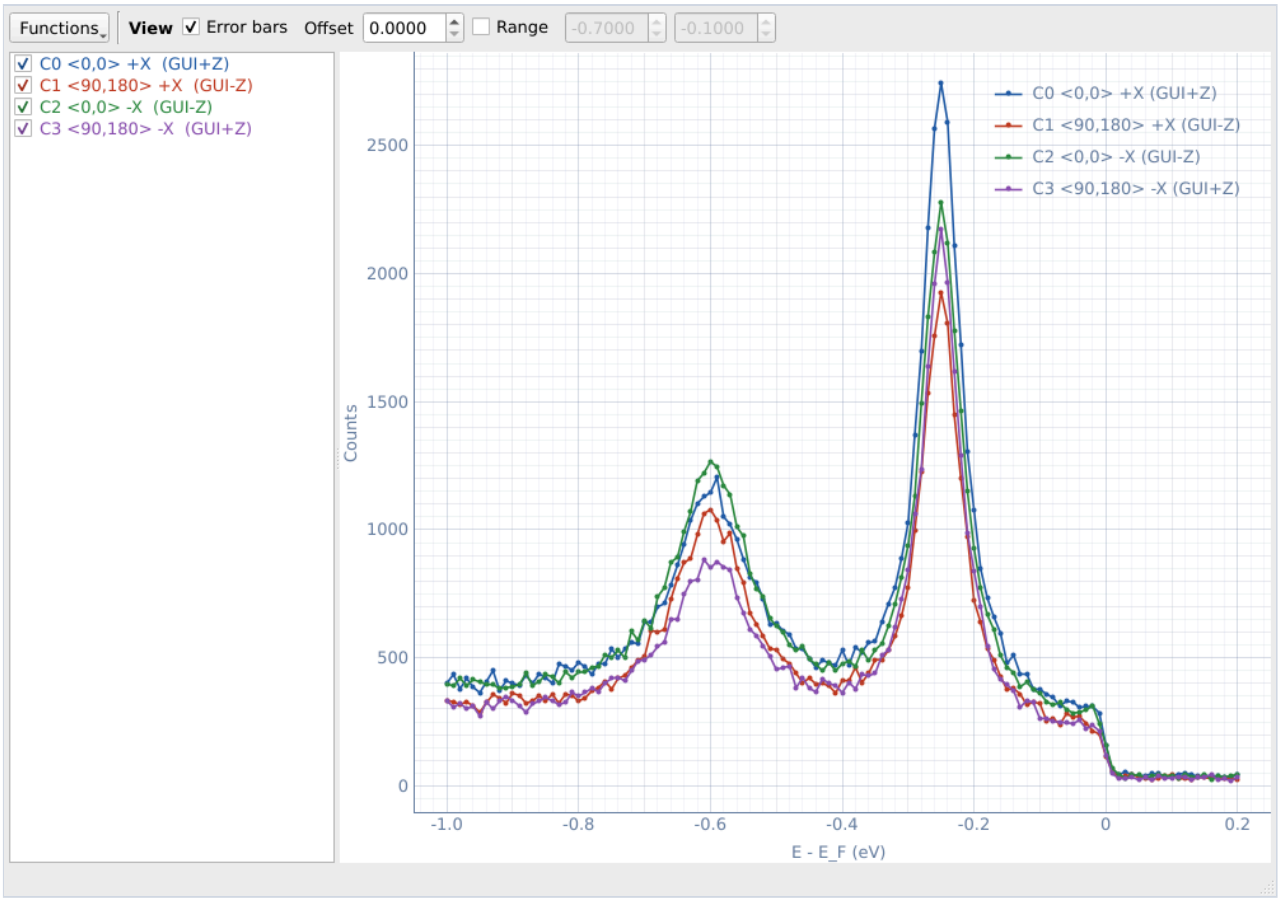
Functions → Curve fit…: **Peaks on a background** (Lorentzian, Gaussian or Voigt peaks placed by clicking or **Find peaks**; none / constant / linear / Shirley background; instrumental resolution; optional **× Fermi edge**), or the **Fermi edge** itself. Weighting is the channel's own σ, √N, or uniform. Every parameter comes with its error; the residual is shown in units of σ.



Fit to list keeps data, fit, residual and components; **Copy results** copies a tab-separated table. A Fermi-edge fit can **List a copy with E_F = 0**.

15.2 Spin analysis

A spin EDC opens with one curve per spin channel:



Functions → Spin analysis…:



- The table reads each channel's label (manipulator setting, target magnetisation, axis and sign counted as "up"); **Axis** and **Counts as** can be overridden.

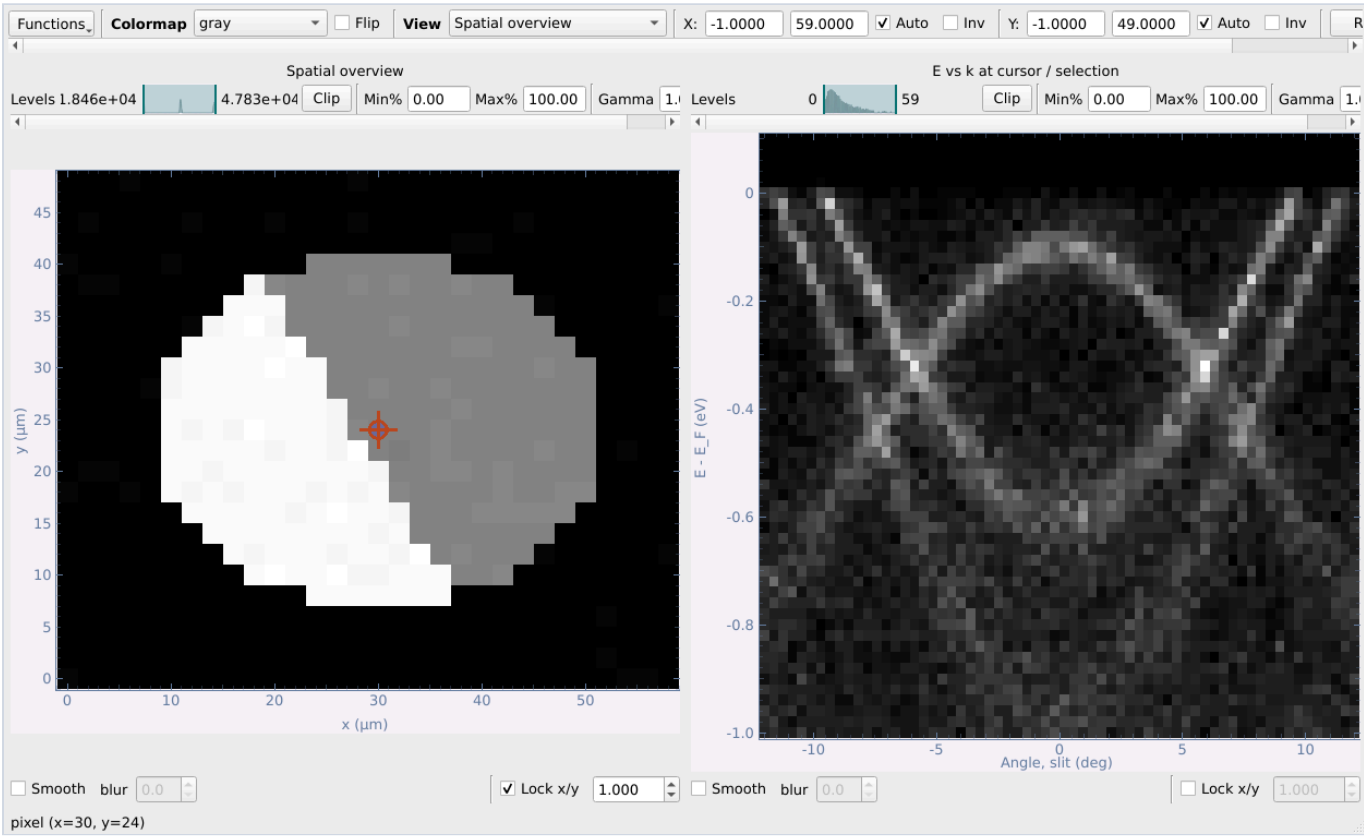
- **Cross ratio** (default when the channels allow it) cancels both the transmission and the target reflectivity exactly; **one pair** is available but carries the instrumental asymmetry, which is measured and reported (ε).
- **S_eff** (Sherman function, default 0.2), **Bin**, optional **Zero reference** window.
- Plots: raw channels, P with its error bars, and I↑ / I↓ with errors.
- **P to list**, **I↑ / I↓ to list**, **As a figure**.

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16. Spatially resolved scans (nano-ARPES)

A real-space scan opens the **spatial map** (left, the total intensity at each position) together with the **E vs k spectrum at the cursor** (right). Drag the red target to read the spectrum at another point. Right-click a panel for the two integrations:

- on the map, **Integrate selection → E vs k panel** sums the spectra over a region of the sample;
- on the spectrum, **Integrate selection → spatial panel** maps the intensity of a chosen (k, E) window across the sample — how a domain or a flake edge is found.

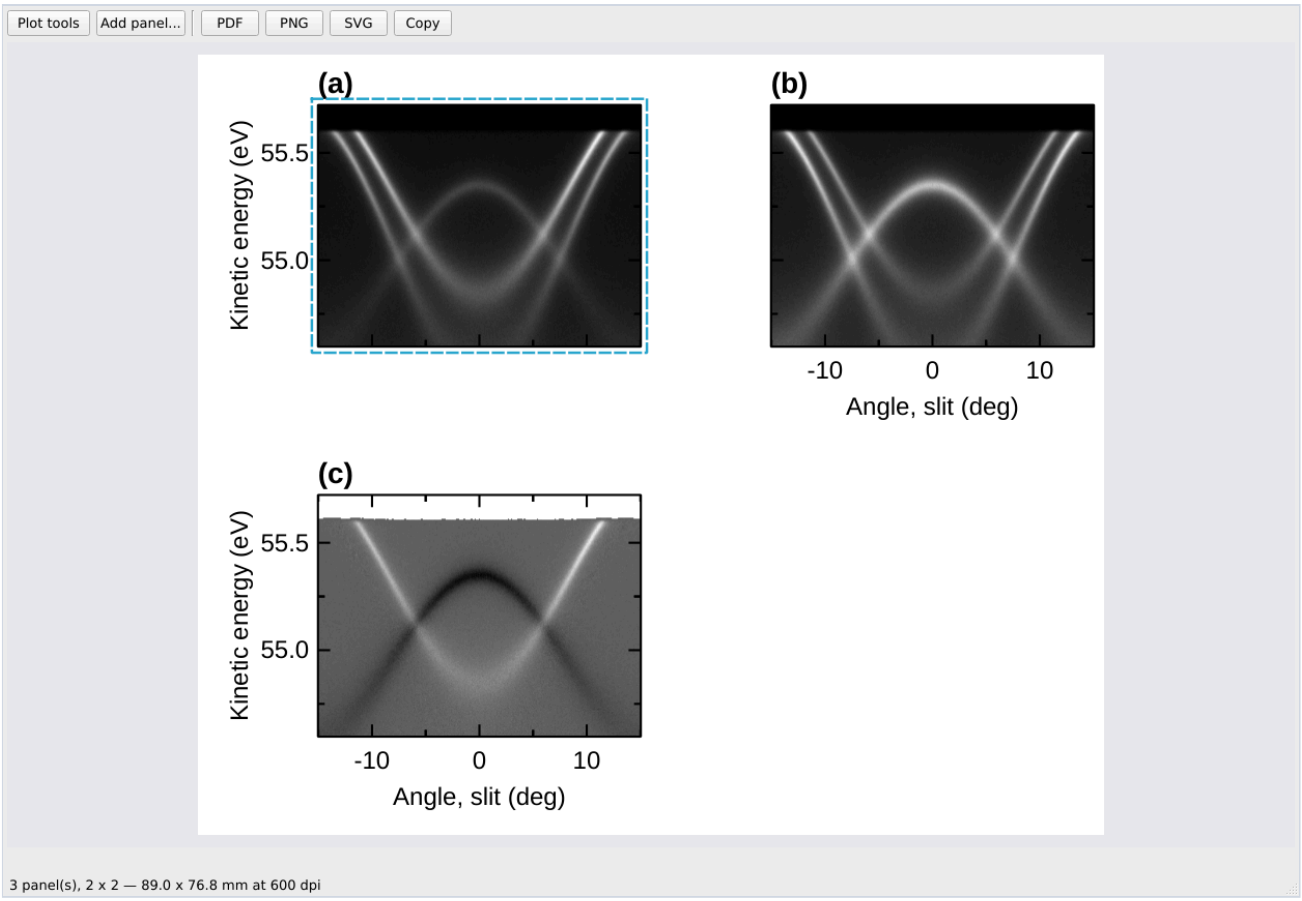


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17. Figures and export

17.1 Figure composer

Select cuts (or curves) in the list → right-click → **Plot as a figure**. The composer lays panels out for a paper: one size per column, labels only on the outer edges, (a)/(b)/(c) letters, shared colour scales if you want them.



Plot tools sets everything (size in mm, fonts, ticks, colour bars, lines, arrows, text, insets, scale bars); **Add panel...** adds another dataset; **PDF**, **PNG**, **SVG** export, **Copy** puts it on the clipboard.

17.2 Slice figure

From a map: *Functions → Slice figure...* lays out a series of slices through the cube — constant-energy contours, or cuts — evenly spaced or at chosen values, with integration, one shared colour scale and titles.

Settings

One panel per slice through the cube. The other two axes become each panel's picture.

Slice: along Kinetic energy (eV)

Positions: evenly spaced / at these values

Number: 6

From / to: 54.1000 — 55.7500

Values: e.g. 0, -0.1, -0.25

Integrate ±: 0.0000

Columns: 3

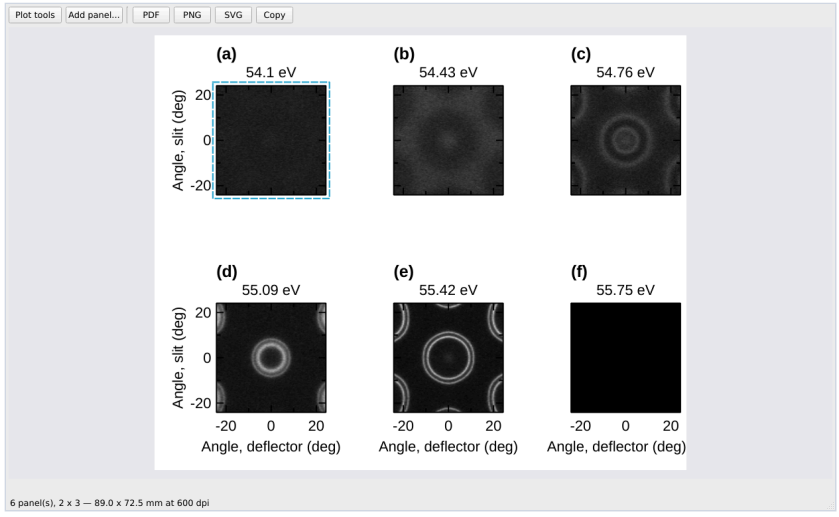
☑ Title each panel with its position

☑ Same colour scale everywhere

6 panel(s); each sums 1 plane(s) of the 166 along Kinetic energy (eV).

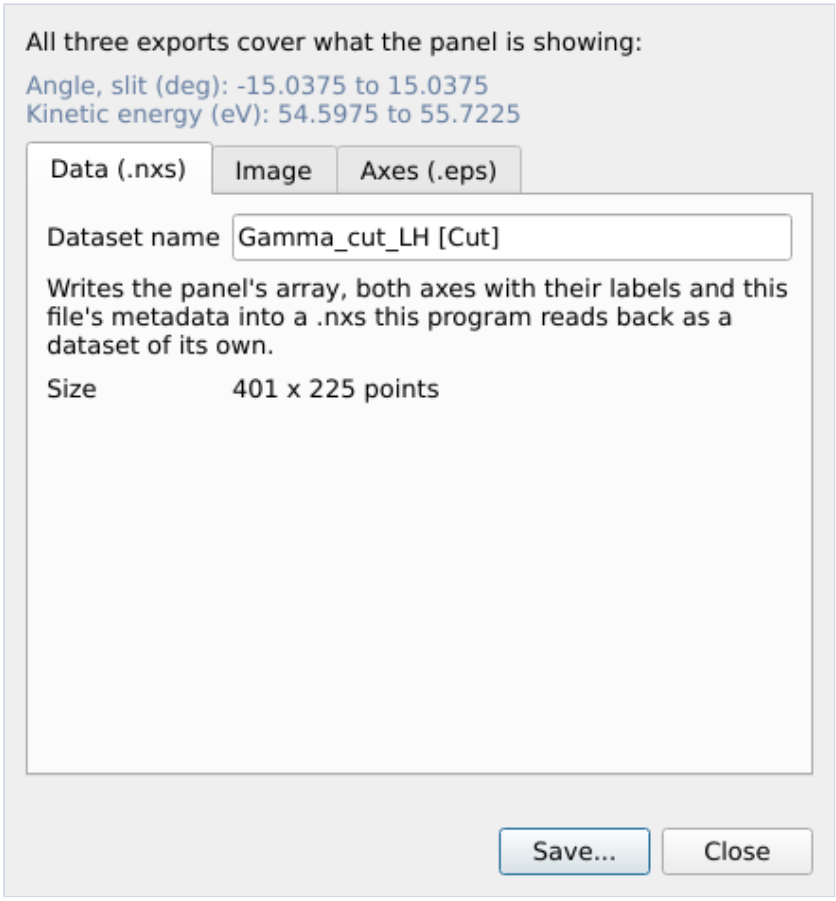
Make the figure | Cancel

Result



17.3 Exporting a panel

Right-click any image → **Export** → **Data (.nxs)...**, **Image...** or **Axes only (.eps)...**. It always exports exactly what the panel shows (range and all): the data as a dataset this program reads back; an image with the colormap applied (PNG, TIFF or JPEG, at a chosen size and dpi, with or without the axes); or the frame, ticks and labels as vector PostScript to place the image in, in Illustrator.



Save (right-click rows in the list) writes datasets to a `.nxs` file of your own; several selected rows go into one file.

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18. Autosave, memory and recovery

- **Autosave.** Every computed dataset (a k-map, a processed cut, a fit result…) is written, uncompressed, to this session's folder in `~/.arpes_viewer/sessions/` the moment it exists. Real-space scans are not (they are the measurement itself, and large).
- **A normal close deletes the autosaved copies.** When you close the main window and confirm (everything you want is saved, or *Close anyway*), the open viewers are closed and the session folder is deleted straight away. A dataset's copy also goes as soon as you save it to a file of your own.
- **Recovery after a crash.** Only if the program did not close normally (a crash, a kill) is the session folder kept: the next start offers its datasets back, and it is deleted after three days.
- **Operations log.** `~/.arpes_viewer/sessions/operations.log`, one line per store / save / discard — how to find what was where.
- **Memory box.** This program's memory, the system's, how many viewers and HDF5 files are open, and how much computed data is held. **Free memory** drops cached copies of computed data (they are re-read from their autosaved file when needed); **Close all viewers** closes every window and

then frees memory.
- **Long operations** (reading a big map, converting, de-gridding, assembling a folder) run off the GUI thread with a progress bar and a working Cancel.
- **Closing** with unsaved computed data asks first, and names the datasets: **Save to a file...**, **Close anyway** (the autosaved copies are deleted) or **Cancel** (nothing happens).

19. Tips and troubleshooting

- **"Not available here" in the Functions menu** — hover over the entry: the tooltip says why (e.g. the MDC/EDC fit needs a cut already in $Å^{-1}$).
- **A k conversion is not offered** — the first axis of the map is not an angle (a photon energy, temperature…). Check *First axis of a map is* in the loader.
- **Cut k conversion of an E − E_F cut** — the energy offset is filled in for you from the Fermi fit; if you shifted the axis some other way, type the kinetic energy of E_F (photon energy − work function) into *Energy offset*.
- **A file is misread** — choose the reader explicitly in the loader, and see *Assumptions worth knowing about* in the README (axis order in SOLEIL files).
- **Dragging the cursor on a big map is sluggish** — a map read lazily from its file re-slices in about 0.1 s per window, so the other windows wait for the mouse to pause. Once the map has been read into memory it follows live.
- **The machine is paging** — watch the Memory bar (amber past a third of the RAM, red past a half); use **Free memory** or **Close all viewers**.